

Stochastic wave equation with time-independent noise

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Abstract

In this article, we study the stochastic wave equation in dimension $d \leq 3$, driven by a Gaussian noise W which does not depend on time. We assume that the covariance function of the noise is of the form $\gamma = K * K$ for some symmetric non-negative function K (satisfying a scaling condition). The solution is interpreted in the Skorohod sense, using Malliavin calculus. The main result gives the exact asymptotic behaviour of the p -th moment of the solution when the time is large.

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1 Introduction

We consider the stochastic wave equation:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) = \Delta u(t, x) + \sqrt{\theta} u(t, x) \dot{W}(x), & t > 0, x \in \mathbb{R}^d \ (d \leq 3) \\ u(0, x) = 1, \quad \frac{\partial u}{\partial t}(0, x) = 0, \end{cases} \quad (1.1) \quad \boxed{\text{E:SWE}}$$

where $\theta > 0$ and $W = \{W(\varphi); \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ is a (time-independent) Gaussian process with mean zero and covariance:

$$E[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}},$$

where μ is a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$. The solution is interpreted in the Skorohod sense, as explained in Section 2 below.

We denote by G the fundamental solution of the wave equation on $\mathbb{R}^d$:

$$G(t, x) = \begin{cases} \frac{1}{2} 1_{\{|x| < t\}} & \text{if } d = 1, \\ \frac{1}{2\pi} \frac{1}{\sqrt{t^2 - |x|^2}} 1_{\{|x| < t\}} & \text{if } d = 2, \\ \frac{1}{4\pi t} \sigma_t & \text{if } d = 3, \end{cases} \quad (1.2) \quad \{?\}$$

where σ_t is the surface measure on the sphere $\{x \in \mathbb{R}^3; |x| = t\}$. Here $|\cdot|$ is the Euclidean norm in $\mathbb{R}^d$. Note that

$$\mathcal{F}G(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|}, \quad \text{for all } \xi \in \mathbb{C}^d \text{ and } d \geq 1, \quad (1.3) \quad \boxed{\text{E:FG}}$$

where $\mathcal{F}\varphi(\xi) = \int_{\mathbb{R}^d} e^{-i\xi \cdot x} \varphi(x) dx$ is the Fourier transform of a function $\varphi \in L^1(\mathbb{R}^d)$.

The goal of this article is to derive sharp solvability conditions and the exact asymptotic behaviour of the p -th moment of the solution to equation (1.1). The heat equation with time-independent noise W as above was also studied in Hu [10] and Hu et al [11], but the investigations related to the wave equation seem to be new in the literature.

It is known that the Fourier transform of μ , denoted by γ , is also nonnegative and nonnegative definite tempered measure. We will need the following assumptions on our noise:

$\langle \text{A:Main} \rangle$ **Assumption A.** Assume that

- (i) both μ and γ are absolutely continuous with respect to the Lebesgue measure; and
- (ii) for some $\alpha \in (0, d)$, γ satisfies the following scaling property:

$$\gamma(cx) = c^{-\alpha} \gamma(x) \quad \text{for all } c > 0, x \in \mathbb{R}^d. \quad (1.4) \quad \boxed{\text{E:scaleGamma}}$$

- (iii) there exists a nonnegative function K on $\mathbb{R}^d$ such that

$$\gamma = K * K, \quad (1.5) \quad \boxed{\text{E:gamma=K*K}}$$

where “ $*$ ” refers to the spatial convolution.

Remark 1.1. Part (i) of Assumption A implies that there exists a nonnegative definite (possibly signed) function K such that the decomposition (1.5) holds. Indeed, let $\varphi(\xi)$ be the density of μ . It is nonnegative and hence $\sqrt{\varphi(\xi)}$ is a well defined function. Viewed as a nonnegative tempered measure, its inverse Fourier transform, denoted as $K := \mathcal{F}^{-1}(\sqrt{\varphi})$, is a nonnegative definite (tempered) measure. Therefore, we have (1.5) in the sense of distribution. Moreover, the absolute continuity of γ entails that K also admits a density. However, existence of a nonnegative K is not immediate. Hence, we list it as part (iii) of the assumption. Part (iii) is always satisfied for all examples that we are interested.

Remark 1.2. The scaling property of γ implies that $\mu(d\xi) = \varphi(\xi)d\xi$ has the following scaling property:

$$\varphi(c\xi) = c^{-(d-\alpha)} \varphi(\xi) \quad \text{or equivalently} \quad \mu(cA) = c^\alpha \mu(A) \quad (1.6) \quad \boxed{\text{E:scaleMu}}$$

for all $c > 0$, $x \in \mathbb{R}^d$ and $A \in \mathcal{B}(\mathbb{R}^d)$.

Assumption A is satisfied by the following examples below and we will treat the white noise case (when $\alpha = d$) separately:

$\langle \text{E:Riesz} \rangle$ **Example 1.3.** $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$. In this case, $\mu(d\xi) = C_{d,\alpha} |\xi|^{-(d-\alpha)} d\xi$ and $K(x) = \beta_{\alpha,d} |x|^{-(d+\alpha)/2}$, where

$$C_{d,\alpha} = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)} \quad \text{and} \quad \beta_{\alpha,d} = \pi^{-d/4} \frac{\Gamma((d+\alpha)/4)}{\Gamma((d-\alpha)/4)} \sqrt{\frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)}}. \quad (1.7) \quad \boxed{\text{E:constCBeta}}$$

The heat equation with this noise was studied in Hu et al [11].

$\langle \text{E:fBM} \rangle$ **Example 1.4.** $\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}$ for some $\alpha_1, \dots, \alpha_d \in (0, 1)$. In this case, $\mu(d\xi) = \prod_{i=1}^d (C_{1,\alpha_i} |\xi_i|^{-(1-\alpha_i)} d\xi_i)$ and $K(x) = \prod_{i=1}^d (\beta_{\alpha_i,1} |x_i|^{-(1+\alpha_i)/2})$, where C_{1,α_i} and $\beta_{\alpha_i,1}$ are given by (1.7). The function γ satisfies the scaling relation (1.4) with $\alpha = \sum_{i=1}^d \alpha_i$. In the parametrization $\alpha_i = 2 - 2H_i$ with $H_i \in (\frac{1}{2}, 1)$, the noise W corresponds to the fractional Brownian sheet with indices $H_1, \dots, H_d$. The heat equation with this noise was studied in Hu [10].

Example 1.5. One can certainly construct examples by combining the previous two examples. For example, one can partition the d coordinates into $k \in \{1, \dots, d\}$ groups with $d_i \geq 1$ coordinates in the i th group such that $d = d_1 + \dots + d_k$. Amount each group of coordinates, one imposes a Riesz kernel with parameter $\alpha_i \in (0, d_i)$, that is, $\gamma(x) = \prod_{i=1}^k |x^{(i)}|^{-\alpha_i}$ where $x^{(i)}$ denote the set of coordinates of x in the i th partition. Then, $\mu(d\xi) = C \prod_{i=1}^k |\xi^{(i)}|^{-(d_i - \alpha_i)}$. The scaling property (1.4) is satisfied with $\alpha = \sum_{i=1}^k \alpha_i \in (0, d)$. In particular, when $k = 1$ (resp. $k = d$), this example reduces to Example 1.3 (resp. 1.4).

Assumption A exclude the white noise case, which will be treated separately in this paper. Formally, Assumption B below corresponds Assumption A for $d = 1, 2$ with $\mu = 1$ (the Lebesgue measure), $\alpha = d$, and $\gamma = K = \delta_0$.

Assumption B. Assume that $d = 1$ or 2 and $\gamma = \delta_0$.

We introduce the following variational quantity: for any nonnegative definite measure f and $\theta > 0$, define

$$\mathcal{M}(f, \theta) = \sup_{g \in \mathcal{F}_d} \left\{ \langle g^2 * f, g^2 \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx \right\}, \quad (1.8) \quad \boxed{\text{E:cm}}$$

where $*$ refers to the convolution, $\mathcal{F}_d = \left\{ g \in W^{1,2}(\mathbb{R}^d); \|g\|_{L^2(\mathbb{R}^d)} = 1 \right\}$ and $W^{1,2}(\mathbb{R}^d)$ is the Sobolev space. We denote $\mathcal{M} = \mathcal{M}(\gamma, 1)$ and $\mathcal{M}(f) = \mathcal{M}(f, 1)$.

Here are the main results of this article.

Theorem 1.6. Under Assumption A with $d \leq 3$, equation (1.1) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$, $t > 0$ and $x \in \mathbb{R}^d$ if and only if $\alpha < 3$. Moreover, if $\alpha < 3$, by setting

$$t_p := (p - 1)^{1/(4-\alpha)} t, \quad (1.9) \quad \boxed{\text{E:tp}}$$

it holds that

$$\lim_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t, x)\|_p = \theta^{\frac{1}{3-\alpha}} \left(\frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}. \quad (1.10) \quad \boxed{\text{E:Main}}$$

In particular, by freezing an arbitrary $p \geq 2$ in (1.10), we have that

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}(|u(t, x)|^p) = 2^{\frac{3}{2} + \frac{5}{2(3-\alpha)}} p [(p-1)\theta]^{\frac{1}{3-\alpha}} (3-\alpha)(4-\alpha)^{-\frac{4-\alpha}{3-\alpha}} \mathcal{M}^{\frac{4-\alpha}{2(3-\alpha)}}, \quad (1.11) \quad \boxed{\text{E:tMain}}$$

and by freezing an arbitrary $t > 0$ in (1.10), we have that

$$\lim_{p \rightarrow \infty} p^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}(|u(t, x)|^p) = 2^{\frac{3}{2} + \frac{5}{2(3-\alpha)}} t^{\frac{4-\alpha}{3-\alpha}} \theta^{\frac{1}{3-\alpha}} (3-\alpha)(4-\alpha)^{-\frac{4-\alpha}{3-\alpha}} \mathcal{M}^{\frac{4-\alpha}{2(3-\alpha)}}. \quad (1.12) \quad \boxed{\text{E:pMain}}$$

The necessary and sufficient condition in Theorem 1.6 is proved in Theorem 2.2. The matching upper and lower bounds in (1.10) are proved in Theorems 5.1 and 6.1, respectively. The two limits (1.11) and 1.12 are direct consequences of (1.10).

We may also discuss the finite time blowup of p -th moments when $d = \alpha = 3$ following ideas of Theorem 3.14 of Chen et al [7].

Theorem 1.7. *Under Assumption B, namely, $\gamma = \delta_0$ and $d = 1$ or 2 , equation (1.1) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$, $t > 0$ and $x \in \mathbb{R}^d$. Moreover, it holds that*

$$\lim_{t_p \rightarrow \infty} t_p^{-\frac{4-d}{3-d}} \log \|u(t, x)\|_p = \theta^{\frac{1}{3-d}} \left(\frac{1}{2} \right)^{\frac{d}{2(3-d)}} \frac{3-d}{2} \left(\frac{2\mathcal{M}(\delta_0)^{1/2}}{4-d} \right)^{\frac{4-d}{3-d}}. \quad (1.13)$$

In particular, by freezing an arbitrary $p \geq 2$ in (1.13), we have that

$$\lim_{t \rightarrow \infty} t^{-\frac{4-d}{3-d}} \log \mathbb{E}(|u(t, x)|^p) = 2^{\frac{3}{2} + \frac{5}{2(3-d)}} p [(p-1)\theta]^{\frac{1}{3-d}} (3-d) (4-d)^{-\frac{4-d}{3-d}} \mathcal{M}(\delta_0)^{\frac{4-d}{2(3-d)}}, \quad (1.14)$$

and by freezing an arbitrary $t > 0$ in (1.13), we have that

$$\lim_{p \rightarrow \infty} p^{-\frac{4-d}{3-d}} \log \mathbb{E}(|u(t, x)|^p) = 2^{\frac{3}{2} + \frac{5}{2(3-d)}} t^{\frac{4-d}{3-d}} \theta^{\frac{1}{3-d}} (3-d) (4-d)^{-\frac{4-d}{3-d}} \mathcal{M}(\delta_0)^{\frac{4-d}{2(3-d)}}. \quad (1.15)$$

I haven't done the proofs for Theorem 1.7. Will do latter.

Remark 1.8. It is known that *Dalang's condition*:

$$\int_{\mathbb{R}^d} \frac{1}{1 + |\xi|^2} \mu(d\xi) < \infty, \quad (1.16)$$

is the necessary and sufficient condition for the existence and uniqueness of the corresponding stochastic heat equation. This is due to the fact that the Fourier transform of the heat kernel is nonnegative. In contrast, the oscillatory nature of the Fourier transform of the wave kernel (see (1.3)) makes (1.16) only a sufficient condition; see (2.9) below. Note that under Assumption A, Dalang's condition (1.16) becomes $\alpha \in (0, 2 \wedge d)$. The existence and uniqueness part of Theorem 1.6 leverages this oscillatory property and hence requires weaker conditions than (1.16). Moreover, the exact asymptotics in (1.11) show that our conditions in Theorem 1.6 are indeed necessary and sufficient conditions for the existence and uniqueness of a solution $u(t, x)$ in $L^p(\Omega)$ for all $t \geq 0$, $x \in \mathbb{R}^d$ and $p \geq 1$.

Remark 1.9. Similarly to Theorem 1.6, under Assumption A, for all $d \geq 1$ and $\alpha \in (0, 2 \wedge d)$, it can be proved that the solution to the heat equation satisfies: for any $p \geq 2$,

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}[u(t, x)^p] = \frac{1}{2} p(p-1)^{\frac{2}{2-\alpha}} \theta^{\frac{2}{2-\alpha}} \frac{2-\alpha}{2} \left(\frac{4\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}. \quad (1.17)$$

Note that under Assumption A, the condition $\alpha \in (0, 2 \wedge d)$ is equivalent to Dalang's condition (1.16). In Theorem 6.5 of Hu et al [11], it is proved that for all $t \geq 0$, $x \in \mathbb{R}^d$ and $p \geq 2$, there exist two constants $C' \geq C > 0$ such that

$$C \exp \left(C t^{\frac{4-\alpha}{2-\alpha}} p^{\frac{4-\alpha}{2-\alpha}} \right) \leq \mathbb{E} [u(t, x)^p] \leq C' \exp \left(C' t^{\frac{4-\alpha}{2-\alpha}} p^{\frac{4-\alpha}{2-\alpha}} \right).$$

Eq. (1.17) closes the gap between these two constants.

Remark 1.10. The following representation plays a fundamental role in this paper thanks to the decomposition (1.5):

$$\begin{aligned} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \int_{[0, t]_<^n} \int_{(\mathbb{R}^d)^n} \right. \\ &\quad \left. \times \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) dy ds \right]^2 d\mathbf{x}, \end{aligned} \quad (1.18) \quad \boxed{\text{E:keyNorm}}$$

where $\tilde{f}_n(\cdot, x; t)$ is the kernel appearing in the Wiener chaos expansion of the solution, Σ_n is the set of permutations of $1, \dots, n$, and we use convention that $s_0 = 0$, and $y_{\sigma(0)} = 0$. Note that the integral in (1.18) should be properly interpreted, which will be done latter.

Throughout the article, for any $p > 0$, we use $\|\cdot\|_p$ to denote the $L^p(\Omega)$ -norm. The letter α is reserved for the scaling index in (1.4) and β for

$$\beta := \frac{4 - \alpha}{3 - \alpha}. \quad (1.19) \quad \boxed{\text{E:beta}}$$

In any dimension $d \geq 1$, the Fourier transform of $G(t, \cdot)$ has the scaling property:

$$\mathcal{F}G(t, \cdot)(c\xi) = \frac{1}{c} \mathcal{F}G(ct, \cdot)(\xi) \quad \text{for any } c > 0. \quad (1.20) \quad \boxed{\text{E:scaleFG}}$$

This article is organized as follows. In Section 2, we introduce some basic elements of Malliavin calculus (see Nualart [16]) and set some notation for the rest part of the paper. In passing, we present a sufficient condition for the existence and uniqueness of the solution, which doesn't impose any scaling properties of the correlation function γ . The necessary and sufficient condition for the existence and uniqueness of a solution under Assumption A for $d \leq 3$ is presented in Section 3. The asymptotics for $p = 2$ in (1.11) is proved in Section 4. The upper and lower bounds for the asymptotics in (1.11) are proved in Sections 5 and 6, respectively. The appendices contain some auxiliary results.

2 Some Preliminaries

$\langle \text{S:Pre} \rangle$ In this section, we present some preliminaries and set up some notation.

2.1 Elements of Malliavin calculus

This section needs to be expanded.

Let $\mathcal{H}$ be the completion of $\mathcal{D}(\mathbb{R}^d)$ with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$. The map $\varphi \mapsto W(\varphi)$ is an isometry from $\mathcal{D}(\mathbb{R}^d)$ to $L^2(\Omega)$, which can be extended to $\mathcal{H}$. Then $\{W(\varphi); \varphi \in \mathcal{H}\}$ is an isonormal Gaussian process, i.e. $E[W(\varphi)W(\psi)] = \langle \varphi, \psi \rangle_{\mathcal{H}}$ for any $\varphi, \psi \in \mathcal{H}$.

We denote by δ the Skorohod integral with respect to W , and by $\text{Dom } \delta$ its domain. For any $u \in \text{Dom } \delta$, we write

$$\delta(u) = \int_{\mathbb{R}^d} u(x)W(\delta x).$$

By definition, $\delta(u)$ is a random variable in $L^2(\Omega)$.

Definition 2.1. A random field $u = \{u(t, x); t \geq 0, x \in \mathbb{R}^d\}$ such that $\|u(t, x)\|_2 < \infty$ for all $t \geq 0$ and $x \in \mathbb{R}^d$ is called a *mild solution* to equation (1.1) if the random field $\{G(t-s, x-y)u(s, y) : s \in [0, t], y \in \mathbb{R}^d\}$ is Skorohod integrable for each $s \in [0, t]$ fixed and satisfies the following stochastic integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y)u(s, y)W(\delta y) \right) ds. \quad (2.1)$$

Intuitively, if the solution exists, it should be given by the series:

$$u(t, x) = 1 + \sum_{n \geq 1} \theta^{n/2} I_n(f_n(\cdot, x; t)), \quad (2.2)$$

where I_n is the multiple integral with respect to W . If $d \leq 2$, the kernel $f_n(\cdot, x; t)$ is an integrable function on $(\mathbb{R}^d)^n$ given by

$$f_n(x_1, \dots, x_n, x; t) = \int_{[0, t]_{<}^n} G(t-t_n, x-x_n) \dots G(t_2-t_1, x_2-x_1) d\mathbf{t}, \quad (2.3)$$

where $[0, t]_{<}^n = \{(t_1, \dots, t_n) \in [0, t]^n; t_1 < \dots < t_n\}$. If $d = 3$, $f_n(\cdot, x; t)$ is a finite measure on $(\mathbb{R}^3)^n$, given by:

$$f_n(\cdot, x; t) = \int_{[0, t]_{<}^n} G(t-t_n, x-dx_n) G(t_n-t_{n-1}, x_n-dx_{n-1}) \dots G(t_2-t_1, x_2-dx_1) d\mathbf{t}.$$

In any dimension d , by setting $t_{n+1} = t$, the Fourier transform of $f_n(\cdot, x; t)$ is

$$\mathcal{F}f_n(\cdot, x; t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1}-t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) d\mathbf{t}. \quad (2.4)$$

We denote by $\tilde{f}_n(\cdot, x; t)$ the symmetrization of $f_n(\cdot, x; t)$ in the variables $x_1, \dots, x_n$:

$$\begin{aligned} \tilde{f}_n(x_1, \dots, x_n, x; t) &= \frac{1}{n!} \sum_{\rho \in \Sigma_n} f_n(x_{\rho(1)}, \dots, x_{\rho(n)}, x; t) \\ &= \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} G(t-t_n, x-x_{\rho(n)}) \dots G(t_2-t_1, x_{\rho(2)}-x_{\rho(1)}) d\mathbf{t}, \end{aligned}$$

which is function for $d \leq 2$, a measure with compact support for $d = 3$, and a distribution whose Fourier transform can be obtained from (2.4).

The following result gives the existence and uniqueness of solution. Its proof follows by a standard procedure.

Theorem 2.2. *Equation (1.1) has a unique solution if and only if the series in (2.2) converges in $L^2(\Omega)$, which is equivalent to the series in (2.5) converges. In this case, the solution is given by (2.2) with the second moment give by*

$$\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} \theta^n n! \|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2. \quad (2.5) \quad \boxed{\text{E:ExUn2m}}$$

2.2 Some large deviation results

Some large deviation results in Bass et al [3] play an important role in this paper. In particular, by Theorem 2.3, *ibid.*, with $p = 2$, together with the moment representation (2.6) in Lemma 2.2, *ibid.*, we see that if $\alpha < 4$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n) = \log \rho, \quad (2.6) \quad \boxed{\text{BCR0}}$$

where

$$\rho = \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + |\xi + \eta|^2} \sqrt{1 + |\eta|^2}} d\eta \right]^2 \mu(d\xi). \quad (2.7) \quad \boxed{\text{E:rho}}$$

By Theorem 1.5, *ibid.*, with $p = 2$, one can identify $\rho = \Lambda_\gamma^{(4-\alpha)/2}$ with ¹ $\Lambda_\gamma = \mathcal{M}(\gamma, 2)$. The functional $\mathcal{M}(\cdot, \cdot)$ has the following scaling property:

Lemma 2.3. *Let f be an arbitrary nonnegative definite measure such that for some $\alpha \neq 4$, $f(cA) = c^{d-\alpha} f(A)$ for all $c > 0$ and $A \in \mathcal{B}(\mathbb{R}^d)$. In case f has a density, this scaling property becomes $f(cx) = c^{-\alpha} f(x)$ for all $x \in \mathbb{R}^d$. Then*

$$\mathcal{M}(\Theta f, \theta) = \Theta^{\frac{2}{4-\alpha}} \theta^{-\frac{\alpha}{4-\alpha}} \mathcal{M}(f, 1), \quad \text{for all } \Theta > 0 \text{ and } \theta > 0.$$

This lemma is proved under slightly different settings in Lemmas B.2 and B.3 of Balan and Song [1]. We give the proof below for readers' convenience.

Proof. For any $\theta, \Theta > 0$, by the definition of $\mathcal{M}$ in (1.8), we see that

$$\begin{aligned} \mathcal{M}(\Theta f, \theta) &= \sup_{g \in \mathcal{F}_d} \left\{ \langle g^2 * \Theta f, g^2 \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx \right\} \\ &= \theta \sup_{g \in \mathcal{F}_d} \left\{ \frac{\sqrt{\Theta}}{\theta} \langle g^2 * f, g^2 \rangle^{1/2} - \frac{1}{2} \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx \right\}. \end{aligned}$$

¹Note that there is a small error in Theorem 1.5 of Bass et al [3] which states that $\rho = (2\pi)^{-d} \Lambda_\alpha^{1-\alpha/2}$. This error is due to the fact that in the first line of equation (7.22), *ibid.*, one should have $(2\pi)^{-dp}$ instead of $(2\pi)^{-d(p+1)}$, since $\mathcal{F}(f^p) = (2\pi)^{-d(p-1)} (\mathcal{F}f)^{*p}$.

Let $g \in W^{1,2}(\mathbb{R}^d)$. For any $B > 0$, by defining $\tilde{g}(x) = B^{-d/2}g(B^{-1}x)$, one sees that the $\|\cdot\|_{L^2(\mathbb{R}^d)}$ norm is preserved, i.e., $\|g\|_{L^2(\mathbb{R}^d)} = \|\tilde{g}\|_{L^2(\mathbb{R}^d)}$. Notice that $\mathcal{F}\tilde{g}(\xi) = B^{d/2}\mathcal{F}g(B\xi)$, which implies that $\int_{\mathbb{R}^d} |\xi|^2 |\mathcal{F}\tilde{g}(\xi)|^2 d\xi = B^{-2} \int_{\mathbb{R}^d} |\xi|^2 |\mathcal{F}g(\xi)|^2 d\xi$, or equivalently,

$$\int_{\mathbb{R}^d} |\nabla \tilde{g}(x)|^2 dx = B^{-2} \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx.$$

By the scaling property of f , we see that $(\tilde{g}^2 * f)(x) = B^{-\alpha}(g^2 * f)(B^{-1}x)$ and hence $\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{-\alpha} \langle g^2 * f, g^2 \rangle$. Combining these relations, we see that

$$\mathcal{M}(\Theta f, \theta) = \theta \sup_{\tilde{g} \in \mathcal{F}_d} \left\{ \frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} \langle \tilde{g}^2 * f, \tilde{g}^2 \rangle^{1/2} - \frac{B^2}{2} \int_{\mathbb{R}^d} |\nabla \tilde{g}(x)|^2 dx \right\}.$$

Hence, in order to preserve some scaling property of $\mathcal{M}$, one has to have

$$\frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} = B^2 \iff B = \left(\frac{\sqrt{\Theta}}{\theta} \right)^{\frac{2}{4-\alpha}} \text{ and } \alpha \neq 4.$$

Plugging the above choice of B proves the lemma. $\square$

By the scaling property of $\mathcal{M}$, namely, $\mathcal{M}(\gamma, 2) = (1/2)^{\frac{\alpha}{4-\alpha}} \mathcal{M}(\gamma, 1)$ (see Lemma B.3 of Balan and Song [1] for a proof), we have the following important relation:

$$\rho = \left[(1/2)^{\frac{\alpha}{4-\alpha}} \mathcal{M} \right]^{(4-\alpha)/2} = (1/2)^{\alpha/2} \mathcal{M}^{(4-\alpha)/2}. \quad (2.8) \quad \boxed{\text{E:rhoM}}$$

2.3 A sufficient condition for solvability

Before the end of this section, let us explain why Dalang's condition (1.16) only gives a sufficient condition for the existence and uniqueness of the solution by showing the following theorem; see Remark 1.8 for more detail:

$\langle \text{T:Dalang} \rangle$ **Theorem 2.4.** *Assume that the spectral measure μ is nonnegative and nonnegative definite such that Dalang's condition (1.16) is satisfied. Then the series in (2.2) converges in $L^2(\Omega)$. Hence the solution $u(t, x)$ exists and is unique for all $(t, x) \in \mathbb{R}_+ \times \mathbb{R}^d$ and for all dimensions $d \geq 1$. Moreover,*

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) < \infty \text{ for all } p \geq 2 \text{ and } T > 0.$$

Proof. Fix an arbitrary $t > 0$. From (2.4), apply the inequality $|\sum_{i=1}^N a_i|^2 \leq N \sum_{i=1}^N |a_i|^2$ and the Cauchy-Schwarz inequality to see that

$$\left| \mathcal{F}\tilde{f}_n(\cdot, x; t)(\cdot, x; t)(\xi_1, \dots, \xi_n) \right|^2 \leq \frac{t^n}{(n!)^2} \sum_{\rho \in \Sigma_n} \int_{[0,t]^n} \prod_{k=1}^n \left| \mathcal{F}G(t_{k+1} - t_k, \cdot) \left(\sum_{j=1}^k \xi_{\rho(j)} \right) \right|^2 dt. \quad (2.9) \quad \boxed{\text{E:FGsq}}$$

Using the change of variable $\xi'_j = \xi_{\rho(j)}$ for $j = 1, \dots, n$, we obtain:

$$\begin{aligned} \|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2 &= \int_{(\mathbb{R}^d)^n} |\mathcal{F}\tilde{f}_n(\cdot, x; t)(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &\leq \frac{t^n}{n!} \int_{[0, t]_{<}^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left| \mathcal{F}G(t_{k+1} - t_k, \cdot) \left(\sum_{j=1}^k \xi_j \right) \right|^2 \mu(d\xi_1) \dots \mu(d\xi_n) dt \\ &\leq \frac{t^n}{n!} \int_{[0, t]_{<}^n} \prod_{k=1}^n \left(\sup_{\eta \in \mathbb{R}^d} \int_{\mathbb{R}^d} |\mathcal{F}G(t_{k+1} - t_k, \cdot)(\xi + \eta)|^2 \mu(d\xi) \right) dt. \end{aligned}$$

By Lemma 4.2 of Balan and Song [2], for any $t > 0$,

$$\sup_{\eta \in \mathbb{R}^d} \int_{\mathbb{R}^d} |\mathcal{F}G(t, \cdot)(\xi + \eta)|^2 \mu(d\xi) \leq 2(t^2 \vee 1) \int_{\mathbb{R}^d} \frac{1}{1 + |\xi|^2} \mu(d\xi) =: D_t C_\mu < \infty,$$

where $D_t = 2(t^2 \wedge 1)$ and $C_\mu = \int_{\mathbb{R}^d} \frac{\mu(d\xi)}{1 + |\xi|^2}$. Hence, $\|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2 \leq \left(\frac{t^n}{n!}\right)^2 (D_t C_\mu)^n$ and

$$\sum_{\geq 0} \theta^n \mathbb{E}(|I_n(f_n(\cdot, x; t))|^2) = \sum_{n \geq 1} \theta^n n! \|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2 \leq \sum_{n \geq 1} \theta^n \frac{t^{2n}}{n!} D_t^n C_\mu^n < \infty.$$

Finally, by Minkowski's inequality and the hypercontractivity property (see Theorem B.1),

$$\begin{aligned} \|u(t, x)\|_p &\leq \sum_{n \geq 0} \theta^{n/2} \|I_n(f_n(\cdot, x; t))\|_p \leq \sum_{n \geq 0} \theta^{n/2} (p-1)^{n/2} \|I_n(f_n(\cdot, x; t))\|_2 \\ &= \sum_{n \geq 0} \theta^{n/2} (p-1)^{n/2} (n!)^{1/2} \|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}} \leq \sum_{n \geq 0} \frac{\theta^{n/2} (p-1)^{n/2} t^n}{(n!)^{1/2}} (D_t C_\mu)^{n/2} < \infty. \end{aligned}$$

This completes the proof of Theorem 2.4. $\square$

3 Solvability: the necessary and sufficient condition for $d \leq 3$

$\langle \text{S:alpha3} \rangle$ The aim of this section is to prove the following theorem:

$\langle \text{T:iff} \rangle$ **Theorem 3.1.** *Suppose that Assumption A holds and suppose that $d \leq 3$. Equation (1.1) has a unique solution $u(t, x)$ for all $t > 0$ and $x \in \mathbb{R}^d$ if and only if $\alpha < 3$.*

We need some preparations before proving this theorem. Let us first introduce some notation. For $d \geq 2$, set

$$L_n(y_1, \dots, y_n) = \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0, t]_{<}^n} \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) ds dt, \quad \text{and} \quad (3.1) \quad \text{?E:Ln?}$$

$$H_n(t, \mathbf{x}) = \sum_{\sigma \in \Sigma_n} \int_{[0, t]_{<}^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) dy ds, \quad (3.2) \quad \text{E:Hn}$$

and for $d = 3$ and for any $\varepsilon > 0$, set

$$L_{n,\varepsilon}(y_1, \dots, y_n) = \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \prod_{k=1}^n G_\varepsilon(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) ds dt, \quad \text{and} \quad (3.3) \quad \text{?E:Lne?}$$

$$H_{n,\varepsilon}(t, \mathbf{x}) = \sum_{\sigma \in \Sigma_n} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G_\varepsilon(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\mathbf{y} ds, \quad (3.4) \quad \boxed{\text{E:Hne}}$$

where $G_\varepsilon(t, \cdot) = G(t, \cdot) * p_\varepsilon$ with $p_\varepsilon(x) = (2\pi\varepsilon)^{-d/2} e^{-|x|^2/(2\varepsilon)}$.

Remark 3.2. A key observation is that the mollification $G_\varepsilon(t, x)$ is a nonnegative function if and only if $d \leq 3$. Since G is a function when $d \leq 2$, there is no need for such mollification. We only use this mollification for the case $d = 3$, where G is a locally finite measure on $\mathbb{R}^3$ that is singular with respect to Lebesgue's measure. Because the fact that nonnegativity property ceases to be true for $d \geq 4$, the current work can only cover the cases with $d \leq 3$.

Denote

$$L(x) = \int_0^\infty e^{-t} G(t, x) dx \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + |\xi|^2},$$

and

$$L_\varepsilon(x) = \int_0^\infty e^{-t} G_\varepsilon(t, x) dx \quad \text{and} \quad \mathcal{F}L_\varepsilon(\xi) = \int_0^\infty e^{-t} \mathcal{F}G_\varepsilon(t, \cdot)(\xi) dt = \frac{e^{-(\varepsilon/2)|\xi|^2}}{1 + |\xi|^2}.$$

By change of variables, one sees that the Laplace transform of convolutions (in time) becomes product:

$$\begin{aligned} L_n(y_1, \dots, y_n) &= \sum_{\sigma \in S_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \dots L(y_{\sigma(n)} - y_{\sigma(n-1)}), \quad d \leq 2, \\ L_{n,\varepsilon}(y_1, \dots, y_n) &= \sum_{\sigma \in S_n} L_\varepsilon(y_{\sigma(1)}) L_\varepsilon(y_{\sigma(2)} - y_{\sigma(1)}) \dots L_\varepsilon(y_{\sigma(n)} - y_{\sigma(n-1)}), \quad d = 3. \end{aligned}$$

Hence,

$$\begin{aligned} \mathcal{F}L_n(\xi_1, \dots, \xi_n) &= \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2}, \quad d \leq 2, \\ \mathcal{F}L_{n,\varepsilon}(\xi_1, \dots, \xi_n) &= \sum_{\sigma \in \Sigma_n} e^{-\frac{\varepsilon}{2} \sum_{k=1}^n |\xi_{\sigma(k)} + \dots + \xi_{\sigma(n)}|^2} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2}, \quad d = 3. \end{aligned} \quad (3.5) \quad \boxed{\text{E:FLn}}$$

Lemma 3.3. *It holds that*

$$\int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \mathbf{x}) dt \right]^2 d\mathbf{x} = \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n), \quad d \leq 2,$$

and the same relation holds with H_n , $\mathcal{F}L_n$ and $d \leq 2$ replaced by $H_{n,\varepsilon}$, $\mathcal{F}L_{n,\varepsilon}$ and $d = 3$, respectively.

Proof. It suffices to prove the case when $d \leq 2$. Notice that by Fubini's theorem,

$$\begin{aligned} \int_0^\infty e^{-t} H_n(t, \mathbf{x}) dt &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) dy ds dt \\ &= \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) L_n(y_1, \dots, y_n) dy. \end{aligned}$$

Hence,

$$\begin{aligned} \int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \mathbf{x}) dt \right]^2 d\mathbf{x} &= \int_{(\mathbb{R}^d)^n} \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) L_n(y_1, \dots, y_n) dy \right]^2 d\mathbf{x} \\ &= \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n), \end{aligned}$$

where the last step is due to the Plancherel theorem. $\square$

Next we prove the key expression (1.18) which was announced in the introduction.

Lemma 3.4. *If $d \leq 2$ and $\gamma = K * K$ for a symmetric non-negative function K on $\mathbb{R}^d$, then*

$$\|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \mathbf{x}) d\mathbf{x}. \quad (3.6) \quad \boxed{\text{norm-fn-X}}$$

Proof. Using the change of variables $t - t_n = s_1, \dots, t - t_1 = s_n$ in (2.3), we write

$$f_n(x_1, \dots, x_n, x; t) = \int_{[0,t]_{<}^n} G(s_1, x - x_n) \dots G(s_n - s_{n-1}, x_2 - x_1) ds_1 \dots ds_n.$$

Hence,

$$\tilde{f}_n(x_1, \dots, x_n, x; t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0,t]_{<}^n} G(s_1, x - x_{\rho(n)}) \dots G(s_n - s_{n-1}, x_{\rho(2)} - x_{\rho(1)}) ds_1 \dots ds_n.$$

By Lemma 5.6 of [12], (Raluca, isn't the following relation from the definition of the space $\mathcal{H}$? If we do want to refer this result, we need to check the conditions. We do not have the same assumptions as Lemma 5.6 of [12].)

$$\|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2 = \int_{(\mathbb{R}^d)^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \gamma(x_k - y_k) \tilde{f}_n(x_1, \dots, x_n, x; t) \tilde{f}_n(y_1, \dots, y_n, x; t) d\mathbf{x} dy.$$

We obtain:

$$\begin{aligned} \|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2 &= \int_{(\mathbb{R}^d)^n} d\mathbf{x} \int_{(\mathbb{R}^d)^n} dy \prod_{k=1}^n \gamma(x_k - y_k) \\ &\quad \times \left(\frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0,t]_{<}^n} G(s_1, x - x_{\rho(n)}) \dots G(s_n - s_{n-1}, x_{\rho(2)} - x_{\rho(1)}) ds \right) \\ &\quad \times \left(\frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0,t]_{<}^n} G(r_1, x - y_{\sigma(n)}) \dots G(r_n - r_{n-1}, y_{\sigma(2)} - y_{\sigma(1)}) dr \right). \end{aligned}$$

Using the fact that $\gamma(x_k - y_y) = \int_{\mathbb{R}^d} K(x_k - z_k)K(y_k - z_k)dz_k$, we see that:

$$\begin{aligned} \|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} d\mathbf{z} \left[\sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} \int_{(\mathbb{R}^d)^n} \right. \\ &\quad \left. \times \prod_{k=1}^n K(x_k - z_k) G(s_1, x - x_{\rho(n)}) \dots G(s_n - s_{n-1}, x_{\rho(2)} - x_{\rho(1)}) d\mathbf{x} ds \right]^2. \end{aligned}$$

We take $x = 0$ and we use the change of variable $y_i = -x_i$ for $i = 1, \dots, n$. Then,

$$\begin{aligned} &\sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - z_k) G(s_1, 0 - x_{\rho(n)}) \dots G(s_n - s_{n-1}, x_{\rho(2)} - x_{\rho(1)}) d\mathbf{x} ds \\ &= \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(-y_k - z_k) G(s_1, y_{\rho(n)}) \dots G(s_n - s_{n-1}, y_{\rho(2)} - y_{\rho(1)}) dy ds \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0, t]_{<}^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(-y_k - z_k) G(s_1, y_{\sigma(1)}) \dots G(s_n - s_{n-1}, y_{\sigma(n)} - y_{\sigma(n-1)}) dy ds, \end{aligned}$$

where for the last equality we passed to the permutation σ given by $\sigma(k) = \rho(n+1-k)$. We take the square of the last expression and we integrate $d\mathbf{z}$. The conclusion follows using the change of variable $x_k = -z_k$. $\square$

For any measurable function $f : [0, \infty) \rightarrow \mathbb{R}$, by the Cauchy-Schwarz inequality,

$$\left(\int_0^\infty e^{-t} f(t) dt \right)^2 \leq \int_0^\infty e^{-t} f^2(t) dt. \quad (3.7) \quad \boxed{\text{CS-ineq}}$$

Moreover, if f is a non-decreasing function on $[0, \infty)$, then the following *reverse Cauchy-Schwarz inequality* holds, which plays a key role in our development.

$\langle \text{L:NonDecr} \rangle$ **Lemma 3.5.** *If $f : [0, \infty) \rightarrow \mathbb{R}$ is a non-decreasing function, then*

$$2 \int_0^\infty e^{-2t} f^2(t) dt \leq \left(\int_0^\infty e^{-t} f(t) dt \right)^2.$$

Proof. Let τ and τ' be two independent random variables with exponential distribution of mean 1. Then $\tau \wedge \tau'$ has an exponential distribution with density $g(t) = 2e^{-2t}$ and

$$\left(\int_0^\infty e^{-t} f(t) dt \right)^2 = E[f(\tau)]E[f(\tau')] = E[f(\tau)f(\tau')] \geq E[f^2(\tau \wedge \tau')] = 2 \int_0^\infty e^{-2t} f^2(t) dt.$$

$\square$

$\langle \text{L:alpha}<4 \rangle$ **Lemma 3.6.** *If Assumption A holds with $0 < \alpha < d \leq 3$, then*

$$\begin{aligned} &\int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt \\ &\leq 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n), \end{aligned} \quad (3.8) \quad \boxed{\text{UB-Laplace-norm}}$$

and consequently,

$$\int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt \leq (2^{4-\alpha} C'_\mu)^n, \quad (3.9) \text{ ?UB-Laplace-norm}$$

where

$$C'_\mu := \int_{\mathbb{R}^d} \left(\frac{1}{1 + |\xi|^2} \right)^2 \mu(d\xi) < \infty, \quad (\text{or equivalently, } \alpha < 4). \quad (3.10) \text{ E:alpha<4}$$

Proof. It suffices to prove (3.8). This will imply that

$$\begin{aligned} \int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &\leq 2^{(4-\alpha)n} \frac{1}{n!} \int_{(\mathbb{R}^d)^n} \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \left(\frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right)^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &= 2^{(4-\alpha)n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + |\sum_{j=k}^n \xi_j|^2} \right)^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &\leq 2^{(4-\alpha)n} \left[\sup_{\eta \in \mathbb{R}^d} \int_{\mathbb{R}^d} \left(\frac{1}{1 + |\xi + \eta|^2} \right)^2 \mu(d\xi) \right]^n \\ &= 2^{(4-\alpha)n} \left[\int_{\mathbb{R}^d} \left(\frac{1}{1 + |\xi|^2} \right)^2 \mu(d\xi) \right]^n, \end{aligned} \quad (3.11) \text{ E_:alpha<4}$$

where for the last line, we used Lemma 4.1 of Balan and Song [2] or Lemma 3.1 of Chen [6].

To prove (3.8), we treat separately the cases $d \leq 2$ and $d = 3$.

Case 1. Assume that $d \leq 2$. By the scaling property of $\mathcal{F}G$ in (1.20), we see that

$$\mathcal{F}\tilde{f}_n(\cdot, 0; ct)(\xi_1, \dots, \xi_n) = c^{2n} \mathcal{F}\tilde{f}_n(\cdot, 0; t)(c\xi_1, \dots, c\xi_n), \quad \text{for all } c > 0.$$

Using the above scaling property and the scaling property of μ (see (1.6)), we have:

$$\begin{aligned} \int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{(\mathbb{R}^d)^n} |\mathcal{F}\tilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n) dt \\ &= 2^{(4-\alpha)n} \int_0^\infty 2e^{-2t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt \\ &= 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\int_0^\infty 2e^{-2t} H_n^2(t, \mathbf{x}) dt \right) d\mathbf{x}. \end{aligned}$$

where for the last line, we used (3.6). Since the function $t \mapsto H_n(t, \mathbf{x})$ is non-negative and non-decreasing, we can apply Lemma 3.5 to see that

$$\begin{aligned} \int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &\leq 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\int_0^\infty e^{-t} H_n(t, \mathbf{x}) dt \right)^2 d\mathbf{x} \\ &= 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n), \end{aligned}$$

where for the last line we used Lemma 3.3. Relation (3.8) follows using the form of $\mathcal{F}L_n(\xi_1, \dots, \xi_n)$ given by (3.5).

Case 2. Assume that $d = 3$. Consider the kernel

$$f_{n,\varepsilon}(y_1, \dots, y_n, x; t) = \int_{[0,t]_{<}^n} \prod_{k=1}^n G_\varepsilon(t_{k+1} - t_k, y_{k+1} - y_k) d\mathbf{t}, \quad (3.12) \quad \boxed{\text{def-gn}}$$

with $t_{n+1} = t$ and $y_{n+1} = x$. Then

$$\mathcal{F}f_{n,\varepsilon}(\cdot, x; t)(\xi_1, \dots, \xi_n) = \exp\left(-\frac{\varepsilon}{2} \sum_{k=1}^n |\xi_1 + \dots + \xi_k|^2\right) \mathcal{F}f_n(\cdot, x; t)(\xi_1, \dots, \xi_n). \quad (3.13) \quad \boxed{\text{F-g-e?}}$$

If $\tilde{f}_{n,\varepsilon}(\cdot, x; t)$ be the symmetrization of $f_{n,\varepsilon}(\cdot, x; t)$, then

$$\begin{aligned} \mathcal{F}\tilde{f}_{n,\varepsilon}(\cdot, x; t)(\xi_1, \dots, \xi_n) &= \frac{1}{n!} \sum_{\rho \in \Sigma_n} \exp\left(-\frac{\varepsilon}{2} \sum_{k=1}^n |\xi_{\rho(1)} + \dots + \xi_{\rho(k)}|^2\right) \\ &\quad \times \mathcal{F}f_n(\cdot, x; t)(\xi_{\rho(1)}, \dots, \xi_{\rho(n)}). \end{aligned}$$

Note that

$$\lim_{\varepsilon \rightarrow 0+} \mathcal{F}\tilde{f}_{n,\varepsilon}(\cdot, x; t)(\xi_1, \dots, \xi_n) = \mathcal{F}\tilde{f}_n(\cdot, x; t)(\xi_1, \dots, \xi_n). \quad (3.14) \quad \boxed{\text{Fe-conv}}$$

The function $\mathcal{F}\tilde{f}_{n,\varepsilon}(\cdot, 0; t)$ has the following scaling property:

$$\mathcal{F}\tilde{f}_{n,\varepsilon}(\cdot, 0; ct)(\xi_1, \dots, \xi_n) = c^{2n} \mathcal{F}\tilde{f}_{n,\varepsilon/c^2}(\cdot, 0; t)(c\xi_1, \dots, c\xi_n), \quad \text{for any } c > 0.$$

This leads to:

$$\|\tilde{f}_{n,\varepsilon}(\cdot, t)\|_{\mathcal{H}^{\otimes n}}^2 = t^{(4-\alpha)n} \|\tilde{f}_{n,\varepsilon/t^2}(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2.$$

Hence

$$\int_0^\infty e^{-t} \|\tilde{f}_{n,\varepsilon}(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt = 2^{(4-\alpha)n} \int_0^\infty 2e^{-2t} \|\tilde{f}_{n,\varepsilon/4}(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt.$$

Similarly to (3.6), (replacing G by G_ε) we have:

$$\|\tilde{f}_{n,\varepsilon}(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_{n,\varepsilon}^2(t, \mathbf{x}) d\mathbf{x}. \quad (3.15) \quad \boxed{\text{norm-fne}}$$

Since the function $t \mapsto H_{n,\varepsilon}(t, \mathbf{x})$ is non-negative and non-decreasing, by the same argument as in Case I, we see that

$$\begin{aligned} \int_0^\infty e^{-t} \|\tilde{f}_{n,\varepsilon}(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &= 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\int_0^\infty 2e^{-2t} H_{n,\varepsilon/4}^2(t, \mathbf{x}) dt \right) d\mathbf{x} \\ &\leq 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\int_0^\infty e^{-t} H_{n,\varepsilon/4}(t, \mathbf{x}) dt \right)^2 d\mathbf{x} \\ &\leq 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_{n,\varepsilon/4}(\xi_1, \dots, \xi_n)|^2 \mu(d\xi) \dots \mu(d\xi_n), \end{aligned}$$

where the last line is due to Lemma 3.3. By bounding the exponential term in the expression of $\mathcal{F}L_{n,\varepsilon}$ in (3.5) from above by 1, we see that

$$\begin{aligned} \int_0^\infty e^{-t} \|\tilde{f}_{n,\varepsilon}(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &\leq 2^{(4-\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &=: 2^{(4-\alpha)n} \frac{1}{(n!)^2} T_n. \end{aligned}$$

Using Fatou's lemma and (3.14),

$$\begin{aligned} \int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{(\mathbb{R}^d)^n} |\mathcal{F}\tilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &\leq \liminf_{\varepsilon \rightarrow 0^+} \int_0^\infty e^{-t} \int_{(\mathbb{R}^d)^n} |\mathcal{F}\tilde{f}_{n,\varepsilon}(\cdot, 0; t)(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &= \liminf_{\varepsilon \rightarrow 0^+} \int_0^\infty e^{-t} \|\tilde{f}_{n,\varepsilon}(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt \leq 2^{(4-\alpha)n} \frac{1}{(n!)^2} T_n. \end{aligned}$$

This completes the proof of the lemma. $\square$

Now we are ready to prove Theorem 3.1.

Proof of Theorem 3.1. By Theorem 2.2, we need to prove that $\sum_{n \geq 0} \theta^n n! \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 < \infty$ if and only if $\alpha < 3$. By the scaling property of $\mathcal{F}G$ in (1.20), we see that

$$\|\tilde{f}_n(\cdot, x; t)\|_{\mathcal{H}^{\otimes n}}^2 = t^{(4-\alpha)n} \|\tilde{f}_n(\cdot, x; 1)\|_{\mathcal{H}^{\otimes n}}^2. \quad (3.16) \quad \boxed{\text{E:scaleTildeFn}}$$

Hence, provided that $(4 - \alpha)n + 1 > 0$ (which is true for all $n \geq 1$ under the current setting),

$$\int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma((4 - \alpha)n + 1) \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2. \quad (3.17) \quad \boxed{\text{E:GammaLapInt}}$$

Suppose first that $\alpha < 3$. Using Lemma 3.6 and Stirling's formula, it follows that

$$\|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \leq \frac{(2^{4-\alpha} C'_\mu)^n}{\Gamma((4 - \alpha)n + 1)} \leq c_1^n \frac{(2^{4-\alpha} C'_\mu)^n}{(n!)^{4-\alpha}}$$

for some constant $c_1 > 0$ depending on α . Using property (3.16) again, we infer that

$$\sum_{n \geq 0} \theta^n n! \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 \leq \sum_{n \geq 0} \theta^n t^{(4-\alpha)n} c_1^n \frac{(2^{4-\alpha} C'_\mu)^n}{(n!)^{3-\alpha}} < \infty.$$

For the converse implication, suppose that $\sum_{n \geq 0} \theta^n n! \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 < \infty$. This implies that $\|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 < \infty$ for any $n \geq 1$. We first notice that if $\alpha \geq 5$, then by the scaling property (3.16), the integral in (3.17) diverges. Hence, α has to be strictly less than 5. Next, we prove that $\alpha < 4$. Suppose by contradiction that $\alpha \in [4, 5)$. On the one hand, for such α , one can always find a $n \geq 1$ such that $(4 - \alpha)n + 1 > 0$ and

hence the integral in (3.17) is finite. On the other hand, we will show that the integral in (3.17) is bounded from below by ∞ , which is a contradiction and hence proves $\alpha < 4$. We consider only the case $d \leq 2$. (The case $d = 3$ is similar, replacing $\tilde{f}_n(\cdot, 0; t)$, $H_n(t, \mathbf{x})$, L_n by $\tilde{f}_{n,\varepsilon}(\cdot, 0; t)$, $H_{n,\varepsilon}(t, \mathbf{x})$, $L_{n,\varepsilon}$, respectively, and taking the limit as $\varepsilon \downarrow 0$.) Using the Cauchy-Schwarz inequality (3.7), followed by Lemma 3.3, we have:

$$\begin{aligned}
\int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &= \frac{1}{(n!)^2} \int_0^\infty e^{-t} \int_{(\mathbb{R}^d)^n} H_n^2(t, \mathbf{x}) d\mathbf{x} dt \\
&\geq \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \mathbf{x}) dt \right]^2 d\mathbf{x} \\
&= \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\
&= \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\
&\geq \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \left(\frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right)^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\
&= \frac{1}{n!} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + |\sum_{j=k}^n \xi_j|^2} \right)^2 \mu(d\xi_1) \dots \mu(d\xi_n).
\end{aligned}$$

By Fubini's theorem, the innermost integral above must be finite, i.e.

$$\int_{\mathbb{R}^d} \left(\frac{1}{1 + |\xi_n|^2} \right)^2 \mu(d\xi_n) < \infty.$$

By the scaling property of μ , this implies that $\alpha < 4$, which is a contradiction.

Since we proved that $\alpha < 4$, we can now invoke Lemma 4.2 below. Hence, there exists a constant $c_2 > 0$ depending on α such that

$$(n!)^{4-\alpha} \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \geq c_2^n \quad \text{for all } n \geq 1.$$

Hence,

$$\infty > \sum_{n \geq 0} \theta^n n! \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 \geq \sum_{n \geq 0} \theta^n t^{(4-\alpha)n} \frac{c_2^n}{(n!)^{3-\alpha}}, \quad \text{for all } t > 0,$$

which implies that $\alpha < 3$. □

Remark 3.7. From the above proof, one can see that when $d = \alpha = 3$, for any $p \geq 2$, there exists a deterministic finite time $T_p > 0$ such that solution $u(t, x)$ exists with finite p th moments up to time T_p . Raluca and Xia, do you think it is worthy to pursue a little this critical case?

Remark 3.8. We suspect that existence and uniqueness of a solution can be extended for $d \leq 2$ to some *rough noise* cases, namely, the cases when $\alpha \in (d, 3)$.

4 The case $p = 2$

$\langle \text{S:p=2} \rangle$ The aim of this section is to prove the following Theorem:

$\langle \text{T:p2} \rangle$ **Theorem 4.1.** *Under Assumption A if $0 < \alpha < d \leq 3$, then for all $x \in \mathbb{R}^d$, it holds that*

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}(|u(t, x)|^2) = \theta^{\frac{1}{3-\alpha}} \left(\frac{1}{2}\right)^{\frac{\alpha}{2(3-\alpha)}} (3-\alpha) \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha}\right)^{\frac{4-\alpha}{3-\alpha}}.$$

We first prove a lemma.

$\langle \text{two-lim} \rangle$ **Lemma 4.2.** *Under Assumption A with $0 < \alpha < d \leq 3$, it holds that*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4-\alpha)n+1) \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[2^{4-\alpha} 2^{-\frac{\alpha}{2}} \mathcal{M}^{\frac{4-\alpha}{2}} \right] \quad \text{and} \quad (4.1) \quad \boxed{\text{lim1}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{4-\alpha} \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{4-\alpha} \right)^{4-\alpha} 2^{-\frac{\alpha}{2}} \mathcal{M}^{\frac{4-\alpha}{2}} \right]. \quad (4.2) \quad \boxed{\text{lim2}}$$

Proof. It suffices to prove (4.1). Relation (4.2) follows from (4.1), using the fact that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{\Gamma(an+1)}{(n!)^a} = a \log a \quad \text{for any } a > 0, \quad (4.3) \quad \boxed{\text{E:StirlingRate}}$$

which is a consequence of Stirling's formula for the Gamma function. We denote

$$T_n := \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n). \quad (4.4) \quad \boxed{\text{E:Tn?}}$$

First, we prove the upper bound in (4.1). By (3.8) and (3.17),

$$\Gamma((4-\alpha)n+1) \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \leq 2^{(4-\alpha)n} \frac{1}{(n!)^2} T_n.$$

Using (2.6), we obtain the desired upper bound:

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4-\alpha)n+1) \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \right) \leq \log 2^{4-\alpha} + \log \rho.$$

Next, we prove the lower bound in (4.1). For this, we will prove below that:

$$\frac{1}{(n!)^2} T_n \leq \Gamma \left(\frac{4-\alpha}{2} n + 1 \right)^2 \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2. \quad (4.5) \quad \boxed{\text{LB-Tn}}$$

Another application of Stirling's formula shows that

$$\Gamma \left(\frac{(4-\alpha)n}{2} + 1 \right)^2 \sim \Gamma((4-\alpha)n+1) 2^{-(4-\alpha)n} C_n,$$

where $C_n = 2^{-1}(4 - \alpha)^{1/2}(2\pi n)^{1/2}$. Using (4.5), we infer that

$$c_\alpha \frac{2^{(4-\alpha)n}}{C_n} \frac{1}{(n!)^2} T_n \leq \Gamma((4 - \alpha)n + 1) \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2,$$

where $c_\alpha > 0$ is a constant depending on α . Using (2.6) and the fact that $\frac{1}{n} \log C_n \rightarrow 0$ as $n \rightarrow \infty$, we obtain the desired lower bound:

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4 - \alpha)n + 1) \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \right) \geq \log 2^{4-\alpha} + \log \rho.$$

Then plugging the formula (2.8) for ρ proves (4.1). Therefore, it remains to prove (4.5). For this, we treat separately the cases $d \leq 2$ and $d = 3$.

Case 1. Assume that $d \leq 2$. For any $t > 0$ and $\tilde{t} > 0$, define

$$J_n(t, \tilde{t}) = \int_{(\mathbb{R}^d)^n} H_n(t, \mathbf{x}) H_n(\tilde{t}, \mathbf{x}) d\mathbf{x},$$

where $H_n(t, x)$ is defined in (3.2). With this notation, relation (1.18) becomes:

$$\|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \mathbf{x}) d\mathbf{x} = \frac{1}{(n!)^2} J_n(t, t). \quad (4.6) \text{ ?fn-Jn?}$$

Hence, J_n satisfies the same scaling property as $\|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2$:

$$J_n(t, t) = t^{(4-\alpha)n} J_n(1, 1). \quad (4.7) \text{ scaling-Jn}$$

Let τ and $\tilde{\tau}$ be independent exponential random variables of mean 1. By Lemma 3.3,

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tilde{\tau})] &= \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \mathbf{x}) dt \right]^2 d\mathbf{x} \\ &= \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n) = T_n. \end{aligned}$$

On the other hand, by Cauchy-Schwarz inequality and (4.7),

$$\mathbb{E}[J_n(t, \tilde{t})] \leq J_n(t, t)^{1/2} J_n(\tilde{t}, \tilde{t})^{1/2} = t^{(4-\alpha)n/2} \tilde{t}^{(4-\alpha)n/2} J_n(1, 1)$$

for any $t > 0$ and $\tilde{t} > 0$. Hence,

$$\begin{aligned} T_n &= \mathbb{E}[J_n(\tau, \tilde{\tau})] \leq \mathbb{E}[\tau^{(4-\alpha)n/2}] \mathbb{E}[\tilde{\tau}^{(4-\alpha)n/2}] J_n(1, 1) = \Gamma\left(\frac{4-\alpha}{2}n + 1\right)^2 J_n(1, 1) \\ &= \Gamma\left(\frac{4-\alpha}{2}n + 1\right)^2 (n!)^2 \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2. \end{aligned}$$

Case 2. Assume that $d = 3$. As in the proof of Lemma 3.6, we consider the approximation $f_{n,\varepsilon}$ given by (3.12). We define

$$J_{n,\varepsilon}(t, \tilde{t}) = \int_{(\mathbb{R}^d)^n} H_{n,\varepsilon}(t, \mathbf{x}) H_{n,\varepsilon}(\tilde{t}, \mathbf{x}) d\mathbf{x},$$

where $H_{n,\varepsilon}(t, x)$ is defined in (3.4). With this notation, relation (3.15) becomes:

$$\|\tilde{f}_{n,\varepsilon}(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_{n,\varepsilon}^2(t, \mathbf{x}) d\mathbf{x} = \frac{1}{(n!)^2} J_{n,\varepsilon}(t, t).$$

$J_{n,\varepsilon}$ has the same scaling property (4.7) as J_n . As in Case 1,

$$\begin{aligned} \mathbb{E}[J_{n,\varepsilon}(\tau, \tilde{\tau})] &= \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_{n,\varepsilon}(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &= \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \exp\left(-\frac{\varepsilon}{2} \sum_{k=1}^n |\xi_{\sigma(k)} + \dots + \xi_{\sigma(n)}|^2\right) \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^2} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n) \\ &=: T_{n,\varepsilon}. \end{aligned}$$

By the Cauchy-Schwarz inequality and the scaling property of $J_{n,\varepsilon}$, we have:

$$T_{n,\varepsilon} = \mathbb{E}[J_{n,\varepsilon}(\tau, \tilde{\tau})] \leq \Gamma\left(\frac{4-\alpha}{2}n + 1\right)^2 (n!)^2 \|\tilde{f}_{n,\varepsilon}(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2.$$

Relation (4.5) follows letting $\varepsilon \downarrow 0$. □

Proof of Theorem 4.1. By Theorem 3.1, the solution u exists. By the scaling property (3.16), we have:

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n n! \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n n! t^{(4-\alpha)n} \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} \theta^n \frac{t^{(4-\alpha)n}}{(n!)^{3-\alpha}} R_n \quad \text{with } R_n := (n!)^{4-\alpha} \|\tilde{f}_n(\cdot, 0; 1)\|_{\mathcal{H}^{\otimes n}}^2. \end{aligned}$$

We apply Lemma A.2 with $x_n = \theta^n \frac{1}{(n!)^{3-\alpha}}$ and the above R_n . By (4.2), we know that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log R_n = \log R, \quad \text{where } R := \left(\frac{2}{4-\alpha}\right)^{4-\alpha} 2^{-\frac{\alpha}{2}} \mathcal{M}^{\frac{4-\alpha}{2}}. \quad (4.8) \quad \boxed{\text{E:R}}$$

So, for deriving the asymptotic behaviour of $t^{-\beta} \log \mathbb{E}(|u(t, x)|^2)$ (for suitable $\beta > 0$), we can replace R_n by R^n . More precisely, if we can find $\beta > 0$ such that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} \theta^n \frac{1}{(n!)^{3-\alpha}} R^n t^{(4-\alpha)n} = A \quad (4.9) \quad \boxed{\text{E4:A-series}}$$

then

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E} (|u(t, x)|^2) = \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} \theta^n \frac{1}{(n!)^{3-\alpha}} R_n t^{(4-\alpha)n} = A.$$

To find the suitable value β , we use Lemma A.3. More precisely,

$$\begin{aligned} \frac{1}{(\theta R)^{1/(3-\alpha)}} \lim_{t \rightarrow \infty} \frac{1}{t^{(4-\alpha)/(3-\alpha)}} \log \sum_{n \geq 0} \frac{\theta^n R^n t^{(4-\alpha)n}}{(n!)^{3-\alpha}} &= \lim_{t \rightarrow \infty} \frac{1}{(\theta R t^{4-\alpha})^{1/(3-\alpha)}} \log \sum_{n \geq 0} \frac{(\theta R t^{4-\alpha})^n}{(n!)^{3-\alpha}} \\ &= \lim_{t \rightarrow \infty} \frac{1}{t^{1/(3-\alpha)}} \log \sum_{n \geq 0} \frac{t^n}{(n!)^{3-\alpha}} = 3 - \alpha. \end{aligned}$$

This shows that if we choose $\beta = \frac{4-\alpha}{3-\alpha}$ (see (1.19)), then relation (4.9) holds with $A = (3 - \alpha)(\theta R)^{1/(3-\alpha)}$. Summarizing, we obtain:

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E} (|u(t, x)|^2) = (3 - \alpha)(\theta R)^{1/(3-\alpha)}. \quad (4.10) \quad \boxed{\text{E:p2limit}}$$

The conclusion follows using the definition of R in (4.8). $\square$

5 Upper Bound

$\langle \text{S:upper} \rangle$ In this section, we use the $L^2(\Omega)$ asymptotics obtained in Theorem 4.1 and the hypercontractivity to show the upper bounds for the asymptotics stated in the following theorem:

$\langle \text{T:Upper} \rangle$ **Theorem 5.1.** *Under Assumption A with $0 < \alpha < d \leq 3$, it holds that*

$$\limsup_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t, x)\|_p \leq \theta^{\frac{1}{3-\alpha}} \left(\frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}. \quad (5.1) \quad \boxed{\text{E:Upper}}$$

where t_p and $\mathcal{M}$ are defined in (1.9) and (1.8), respectively.

Proof. For any $\lambda > 0$, we denote by u_λ the solution of equation (1.1) with θ replaced by λ . Denote $u = u_\theta$. By L  's moment comparison theorem given by Theorem B.1,

$$\|u(t, x)\|_p \leq \|u_{(p-1)\theta}(t, x)\|_2, \quad \text{for all } p \geq 2.$$

By the scaling property (3.16),

$$\begin{aligned} \|u_{(p-1)\theta}(t, x)\|_2^2 &= \sum_{n \geq 0} n! \theta^n (p-1)^n \left\| \tilde{f}_n(\cdot, x; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} n! \theta^n \left\| \tilde{f}_n \left(\cdot, x; (p-1)^{\frac{1}{4-\alpha}} t \right) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \left\| u \left((p-1)^{1/(4-\alpha)} t, x \right) \right\|_2^2. \end{aligned}$$

This leads to the following important moment inequality:

$$\|u(t, x)\|_p \leq \|u \left((p-1)^{1/(4-\alpha)} t, x \right)\|_2 = \|u(t_p, x)\|_2,$$

which implies that

$$\limsup_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t, x)\|_p \leq \limsup_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t_p, 0)\|_2.$$

Then an application of Theorem 4.1 proves (5.1). $\square$

6 Lower bound

$\langle \text{S:lower} \rangle$ The goal of this section is to prove the following theorem:

$\langle \text{T:Lowbound} \rangle$ **Theorem 6.1.** *Under Assumption A with $0 < \alpha < d \leq 3$, it holds that*

$$\liminf_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t, x)\|_p \geq \theta^{\frac{1}{3-\alpha}} \left(\frac{1}{2}\right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha}\right)^{\frac{4-\alpha}{3-\alpha}}, \quad (6.1) \quad \boxed{\text{E:Low}}$$

where t_p and $\mathcal{M}$ are defined in (1.9) and (1.8), respectively.

We may need to treat the white noise for $d = 1, 2$ separately.

This theorem will be proved at the end of this section. We need first introduce some notation and prove some auxiliary results. For any $\phi \in L^2(\mu)$, let

$$W_n(t, \phi) := \int_{[0, t]_{\leq}^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds, \quad (6.2) \quad \boxed{\text{E:Wnt?}}$$

with $s_0 = 0$. We first give conditions under which $W_n(t, \phi)$ is well defined:

$\langle \text{L:LaplaceW} \rangle$ **Lemma 6.2.** *Under Assumption A with $\alpha < 4$ (see (3.10)), for all $d \geq 1$, $t > 0$ and $\phi \in L^2(\mu)$, $W_n(t, \phi)$ is well defined. Moreover,*

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + |\xi_k + \dots + \xi_n|^2} \mu(d\xi_1) \dots \mu(d\xi_n). \quad (6.3) \quad \boxed{\text{E:LaplaceW}}$$

Proof. Let's first prove the relation (6.3). Denote the left-hand side of (6.3) by I . Then, by Fubini's theorem,

$$\begin{aligned} I &= \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \int_0^\infty e^{-t} \int_{[0, t]_{\leq}^n} \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) ds dt \mu(d\xi_1) \dots \mu(d\xi_n) \\ &= \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \left(\int_0^\infty e^{-u} du \right) \prod_{k=1}^n \left(\int_0^\infty e^{-u_k} \mathcal{F}G(u_k, \cdot)(\xi_k + \dots + \xi_n) du_k \right) \mu(d\xi_1) \dots \mu(d\xi_n), \end{aligned}$$

where for the last equality we used the change of variables $t - s_n = u$ and $s_k - s_{k-1} = u_k$ for $k = 1, \dots, n$. Finally, we observe that $\int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \int_0^\infty e^{-t} \frac{\sin(t|\xi|)}{|\xi|} dt = 1/(1 + |\xi|^2)$ for all $\xi \in \mathbb{C}^d$. This proves (6.3). To see that the right-hand side of (6.3) is finite, one can apply the Cauchy-Schwarz inequality and then use the same arguments as those leading to (3.11). $\square$

Our interest in the quantity $W_n(t, \phi)$ stems from the following proposition.

$\langle \text{LB-1} \rangle$ **Proposition 6.3.** For any $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $1/p + 1/q = 1$, it holds that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2}(q-1)\|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|.$$

As a consequence, the series $|\sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f)|$ converges provided that $\|u(t, 0)\|_p < \infty$, which is the case if Assumption A holds and $\alpha < 3$ (see Theorem 3.1).

Proof. It suffices to prove the inequality. For any distribution $f \in \mathcal{H}$, let

$$Z_f = \exp \left(W(f) - \frac{1}{2}\|f\|_{\mathcal{H}}^2 \right) = \sum_{n \geq 0} \frac{1}{n!} I_n(f^{\otimes n}).$$

Here $f^{\otimes n}$ is the distribution in $\mathcal{S}'(\mathbb{R}^{nd})$ whose Fourier transform is given by: $\mathcal{F}f^{\otimes n}(\xi_1, \dots, \xi_n) = \prod_{k=1}^n \mathcal{F}f(\xi_k)$. If f is a function, then $f^{\otimes n}$ is the function $f^{\otimes n}(x_1, \dots, x_n) = \prod_{k=1}^n f(x_k)$.

Using Hölder's inequality and the fact that $W(f) \sim N(0, \|f\|_{\mathcal{H}}^2)$, we obtain:

$$|\mathbb{E}[u(t, 0)Z_f]| \leq \|u(t, 0)\|_p \|Z_f\|_q = \|u(t, 0)\|_p \exp \left\{ \frac{1}{2}(q-1)\|f\|_{\mathcal{H}}^2 \right\}. \quad (6.4) \quad \boxed{\text{E-uZ-Holder}}$$

By the orthogonality of the Wiener chaos spaces,

$$\mathbb{E}[u(t, 0)Z_f] = \sum_{n \geq 0} \theta^{n/2} \frac{1}{n!} \mathbb{E}[I_n(f_n(\cdot, 0; t))I_n(f^{\otimes n})] = \sum_{n \geq 0} \theta^{n/2} \langle \tilde{f}_n(\cdot, 0; t), f^{\otimes n} \rangle_{\mathcal{H}^{\otimes n}}. \quad (6.5) \quad \boxed{\text{E-uZ}}$$

Using the definition of $\tilde{f}_n(\cdot, 0; t)$, we see that

$$\begin{aligned} \langle \tilde{f}_n(\cdot, 0; t), f^{\otimes n} \rangle_{\mathcal{H}^{\otimes n}} &= \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(\xi_k) \overline{\tilde{f}_n(\cdot, x; t)(\xi_1, \dots, \xi_n)} \mu(d\xi_1) \dots \mu(d\xi_n) \\ &= \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \mathcal{F}G(t_{k+1} - t_k, \cdot)(\xi_1 + \dots + \xi_k) \mu(d\xi_1) \dots \mu(d\xi_n) dt, \end{aligned}$$

with $t_{n+1} = t$. Using the change of variables $s_1 = t - t_n, \dots, s_n = t - t_1$ and $\xi'_1 = \xi_n, \dots, \xi'_n = \xi_1$, we infer that

$$\langle \tilde{f}_n(\cdot, 0; t), f^{\otimes n} \rangle_{\mathcal{H}^{\otimes n}} = W_n(t, \mathcal{F}f). \quad (6.6) \quad \boxed{\text{product-f}}$$

The conclusion follows from (6.4), (6.5) and (6.6). $\square$

Recall that the constant β is defined in (1.19). For any $f \in \mathcal{H}$ and $t > 0$, let f_t be the time-scaled distribution, which is a distribution in $\mathcal{S}'(\mathbb{R}^d)$ whose Fourier transform is

$$\mathcal{F}f_t(\xi) = t^{(2-\alpha)(\beta-1)} \mathcal{F}f(t^{1-\beta}\xi) = t^{\frac{2-\alpha}{3-\alpha}} \mathcal{F}f(t^{1/(\alpha-3)}\xi) \quad \text{for all } \xi \in \mathbb{C}^d. \quad (6.7) \quad \boxed{\text{E:scale-f?}}$$

Note that $f_t \in \mathcal{H}$. If f is a function, then f_t is also a function with the scaling property:

$$f_t(x) = t^{(2-\alpha+d)(\beta-1)} f(t^{\beta-1}x) = t^{\frac{2-\alpha+d}{3-\alpha}} f(t^{1/(3-\alpha)}x), \quad \text{for all } x \in \mathbb{R}^d.$$

We have the following scaling result.

$\langle \text{scaling-ft} \rangle$ **Lemma 6.4.** Under Assumption **A** with $\alpha < 4$, for any $f \in \mathcal{H}$ and $t > 0$, it holds that

$$\|f_t\|_{\mathcal{H}}^2 = t^\beta \|f\|_{\mathcal{H}}^2 \quad \text{and} \quad W_n(t, \mathcal{F}f_t) = W_n(t^\beta, \mathcal{F}f).$$

Proof. Using the change of variable $\xi' = t^{1-\beta}\xi$ and the scaling property of μ , we see that:

$$\|f_t\|_{\mathcal{H}}^2 = t^{2(2-\alpha)(\beta-1)} \int_{\mathbb{R}^d} |\mathcal{F}f(t^{1-\beta}\xi)|^2 \mu(d\xi) = t^{(\beta-1)(4-\alpha)} \|f\|_{\mathcal{H}}^2.$$

The desired relation follows since $(\beta-1)(4-\alpha) = \beta$. The second relation is proved by the change of variable $\xi'_k = t^{1-\beta}\xi_k$, applying the scaling properties of μ and $\mathcal{F}G$ in (1.6) and (1.20), respectively, and then another change of variable $s'_k = t^{\beta-1}s_k$. $\square$

For any $f \in \mathcal{H}$, an application of Proposition 6.3 to f_{t_p} with t_p being defined in (1.9), followed by Lemma 6.4, proves the following proposition:

$\langle \text{P:LB} \rangle$ **Proposition 6.5.** Under Assumption **A** with $\alpha < 3$, for any $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $1/p + 1/q = 1$, it holds that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2}(q-1) t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|. \quad (6.8) \quad \boxed{\text{LB-2}}$$

Note that $W_n(t, \phi)$ may be negative. In order to get rid of the absolute value in (6.8), we need to identify $f \in \mathcal{H}$ for which $W_n(t, \mathcal{F}f)$ is nonnegative. For this purpose, we will need to both restrict our dimensions to $d \leq 3$ and restrict $\mathcal{H}$ to the following space

$$\mathcal{H}_+ := \mathcal{H} \cap C_+(\mathbb{R}^d) \subseteq |\mathcal{H}|,$$

where $C_+(\mathbb{R}^d)$ refers the space of continuous, strictly positive, and nonnegative definite functions on $\mathbb{R}^d$. With these restrictions, by the Plancherel theorem, we see that

$$\begin{aligned} W_n(t, \mathcal{F}f) &= \int_{[0, t]_{<}^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n G(s_k - s_{k-1}, x_k - x_{k-1}) dx ds \\ &=: U_n(t, f), \quad \text{for all } f \in \mathcal{H}_+ \text{ and } d \leq 3, \end{aligned} \quad (6.9) \quad \boxed{\text{E:Untf}}$$

with $s_0 = 0$ and $x_0 = 0$. Since $\prod_{k=1}^n G(s_k - s_{k-1}, x_k - x_{k-1})$ with $(s_1, \dots, s_n)$ fixed is a nonnegative measure with compact support on $(\mathbb{R}^d)^n$, which is true only for $d \leq 3$, and since the other product $(x_1, \dots, x_n) \mapsto \prod_{k=1}^n (f * \gamma)(x_k)$ is a continuous and nonnegative function on $(\mathbb{R}^d)^n$, the dx -integral in (6.9) is finite and nonnegative, i.e.,

$$0 \leq W_n(t, \mathcal{F}f) = U_n(t, f) < \infty, \quad \text{for all } f \in \mathcal{H}_+, n \geq 1, t > 0 \text{ and } d \leq 3. \quad (6.10) \quad \boxed{\text{E:W=U>0}}$$

Now let us define

$$W_n(t) = \sup_{f \in \mathcal{H}; \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}f) \quad \text{and} \quad U_n(t) = \sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=1} U_n(t, f). \quad (6.11) \quad \boxed{\text{E:Wnt?}}$$

Then, when $d \leq 3$, we have that

$$W_n(t) \geq \sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}f) = U_n(t) \geq 0.$$

$\langle \liminf\text{-EW} \rangle$ **Theorem 6.6.** *Under Assumption A with $0 < \alpha < d \leq 3$, if τ is an exponential random variable of mean 1, then*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} [U_n(\tau)] \geq \log \left(2^{-\frac{\alpha}{4}} \mathcal{M}^{\frac{4-\alpha}{4}} \right). \quad (6.12) \quad \boxed{\text{E:EU}}$$

Proof. The proof here follows similar arguments as those in Section 3 of Bass et al [3]. Notice that

$$\mathbb{E} [U_n(\tau)] = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt. \quad (6.13) \quad \boxed{\text{E}_-:\text{EU}}$$

Fix an arbitrary $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = 1$ and set $\phi = \mathcal{F}f$. Since f is nonnegative and nonnegative definite, we see that ϕ is also nonnegative and nonnegative definite (hence symmetric). Moreover, $\|\phi\|_{L^2(\mu)} = \|f\|_{\mathcal{H}} = 1$. By (6.10), we can replace the $U_n(t, f)$ in (6.13) by $W_n(t, \phi)$. Then we apply Lemma 6.2 to the dt integral to see that

$$\mathbb{E} [U_n(\tau)] \geq \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + |\xi_k + \dots + \xi_n|^2} \mu(d\xi_1) \dots \mu(d\xi_n) =: I_n. \quad (6.14) \quad \boxed{\text{E}_-:\text{lb dEU}}$$

Notice that the right-hand side of (6.14) takes the same form as the far right-hand side of Eq. (3.3) of Bass et al [3]. With f and φ in Section 3, ibid., replaced by ϕ and μ , one can apply similar arguments there to show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + |\xi_k + \dots + \xi_n|^2} \mu(d\xi_1) \dots \mu(d\xi_n) = \log \rho(\phi), \quad (6.15) \quad \text{?BCR?}$$

where

$$\rho(\phi) = \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \phi(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + |\xi + \eta|^2} \sqrt{1 + |\eta|^2}} d\eta \right] \mu(d\xi). \quad (6.16) \quad \boxed{\text{E:rhophi}}$$

Hence, we see that any $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = 1$,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} [U_n(\tau)] \geq \log \rho(\mathcal{F}f).$$

We claim that

$$\sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \rho^{1/2}, \quad (6.17) \quad \boxed{\text{E:supHplus}}$$

where ρ is defined in (2.7). Then an application of (2.8) proves (6.12).

It remains to prove (6.17). Notice that for any $h \in L^2(\mathbb{R}^d)$, $\frac{h(\xi)}{\sqrt{1+|\xi|^2}}$ is also in $L^2(\mathbb{R}^d)$. Hence the following function g is well defined:

$$g(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left(\frac{h(\cdot)}{\sqrt{1 + |\cdot|^2}} \right) (x).$$

A first observation is that g is not only in $L^2(\mathbb{R}^d)$ but also in $W^{1,2}(\mathbb{R}^d) \subseteq L^2(\mathbb{R}^d)$ with $\|g\|_{W^{1,2}(\mathbb{R}^d)} = 1$. Indeed,

$$\begin{aligned} \|g\|_{W^{1,2}(\mathbb{R}^d)}^2 &= \|g\|_{L^2(\mathbb{R}^d)}^2 + \|\nabla g\|_{L^2(\mathbb{R}^d)}^2 \\ &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} |\mathcal{F}g(\xi)|^2 d\xi + \frac{1}{(2\pi)^d} \sum_{k=1}^d \int_{\mathbb{R}^d} \xi_k^2 |\mathcal{F}g(\xi)|^2 d\xi \\ &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (1 + |\xi|^2) |\mathcal{F}g(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}^d} h(\xi)^2 d\xi = 1. \end{aligned}$$

Hence, $g \in W^{1,2}(\mathbb{R}^d)$ with $\|g\|_{W^{1,2}(\mathbb{R}^d)} = 1$. Note that by the Sobolev embedding theorem, $g \in C(\mathbb{R}^d)$ provided that $2 > d/2$, i.e., $d < 4$. Hence, the condition $d \leq 3$ implies that $g \in C(\mathbb{R}^d)$.

A second observation is that $\rho(\mathcal{F}f)$ defined in (6.16) can be written as, upon taking the Fourier transforms,

$$\begin{aligned} \rho(\mathcal{F}f) &= \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} ((\mathcal{F}g) * (\mathcal{F}g))(\xi) (\mathcal{F}f)(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} (f * \gamma)(x) g^2(x) dx \\ &= \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} (2\pi)^{-d} \langle f, g^2 \rangle_{\mathcal{H}}. \end{aligned}$$

Then we see that

$$\sup_{f \in \mathcal{H}; \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = (2\pi)^{-d} \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \sup_{f \in \mathcal{H}; \|f\|_{\mathcal{H}}=1} \langle f, g^2 \rangle_{\mathcal{H}} = (2\pi)^{-d} \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \langle g^2, g^2 \rangle_{\mathcal{H}}^{1/2},$$

where the optimal $f \in \mathcal{H}$ can be chosen to be Cg^2 with $C = \|g^2\|_{\mathcal{H}}^{-1}$. Since we have shown that $g \in C(\mathbb{R}^d)$, we see that the supremum in the above equality can be restricted to $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = 1$. In other words,

$$\sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = (2\pi)^{-d} \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \langle g^2, g^2 \rangle_{\mathcal{H}}^{1/2}, \quad (6.18) \quad \boxed{\text{E_rhoFf}}$$

Finally, notice that

$$\begin{aligned} (2\pi)^{-d} \langle g^2, g^2 \rangle_{\mathcal{H}} &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |(\mathcal{F}g^2)(\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |[(\mathcal{F}g) * (\mathcal{F}g)](\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= \left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi + \nu)h(\eta)}{\sqrt{1 + |\xi + \eta|^2} \sqrt{1 + |\eta|^2}} \right]^2 \mu(d\xi) \right)^{1/2}. \end{aligned}$$

Plugging the above identity back to (6.18) proves (6.17). $\square$

The next result gives the scaling property of W_n and U_n .

$\langle \text{scaling-W} \rangle$ **Lemma 6.7.** *Under Assumption A with $\alpha < 4$, we have that $W_n(t) = t^{\frac{4-\alpha}{2}n} W_n(1)$. Moreover, when $d \leq 3$, $U_n(t) = t^{\frac{4-\alpha}{2}n} U_n(1)$.*

Proof. It suffices to prove the case of $W_n(t)$. For any $f \in \mathcal{H}$, denote $\phi = \mathcal{F}f \in L^2(\mu)$. Using the scaling properties of μ and $\mathcal{F}G$ in (1.6) and (1.20), respectively, we see that for any $c > 0$,

$$\|\phi(c^{-1}\cdot)\|_{L^2(\mu)}^2 = c^\alpha \|\phi\|_{L^2(\mu)}^2 \quad \text{and} \quad W_n(t, \phi(c^{-1}\cdot)) = c^{-(2-\alpha)n} W_n(ct, \phi).$$

We use these properties for $c = t^{-1}$. Then $\|t^{\alpha/2}\phi(t\cdot)\|_{L^2(\mu)}^2 = \|\phi\|_{L^2(\mu)}^2$ and $W_n(t, \phi(t\cdot)) = t^{n(2-\alpha)} W_n(1, \phi)$. We multiply the previous relation by $t^{n\alpha/2}$. Using the fact that $W_n(t, c\phi) = c^n W_n(t, \phi)$ for any $c > 0$ and $\phi \in L^2(\mu)$, we obtain: $W_n(t, t^{\alpha/2}\phi(t\cdot)) = t^{\frac{4-\alpha}{2}n} W_n(t, \phi)$. Finally, the lemma is proved by taking the supremum over all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} = 1$. $\square$

$\langle \text{L:asyW1} \rangle$ **Lemma 6.8.** *Under Assumption A with $\alpha < 4$, it holds that*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{\frac{4-\alpha}{2}} W_n(1) \right) \geq \log \left[\left(\frac{2}{4-\alpha} \right)^{\frac{4-\alpha}{2}} 2^{-\frac{\alpha}{4}} \mathcal{M}^{\frac{4-\alpha}{4}} \right].$$

Moreover, when $d \leq 3$, the above bounds holds with $W_n(1)$ replaced by $U_n(1)$.

Proof. It suffices to prove the case for W_n . By the scaling property of W_n given by Lemma 6.7, we have:

$$\mathbb{E}[W_n(\tau)] = \mathbb{E}(\tau^{\frac{4-\alpha}{2}n}) W_n(1) = \Gamma \left(\frac{4-\alpha}{2}n + 1 \right) W_n(1).$$

Using Theorem 6.6, we obtain:

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma \left(\frac{4-\alpha}{2}n + 1 \right) W_n(1) \right) \geq \log \left(2^{-\frac{\alpha}{4}} \mathcal{M}^{\frac{4-\alpha}{4}} \right).$$

An application of (4.3) proves the lemma. $\square$

$\langle \text{L:AsyU} \rangle$ **Lemma 6.9.** *Under Assumption A with $0 < \alpha < d \leq 3$, for all $a, \theta > 0$, there exist two constants $c_1 = c_1(\alpha, \mathcal{M}, a, \theta) > 0$ and $c_2 = c_2(\alpha, \mathcal{M}, a, \theta) > 0$ such that, by setting $n_t = [c_1 t]$ and $n_p = [c_2 p^{1/(4-\alpha)}]$, it holds that*

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log (a^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq (a\sqrt{\theta})^{\frac{2}{4-\alpha}} 2^{-\frac{\alpha}{2(4-\alpha)}} \mathcal{M}^{1/2}. \quad (6.19) \quad \boxed{\text{E:tAsyU}}$$

Proof. Let $\varepsilon > 0$ be arbitrary. By Lemma 6.8, there exists $N_\varepsilon \in \mathbb{N}$ such that for all $n \geq N_\varepsilon$, $(n!)^{\frac{4-\alpha}{2}} U_n(1) \geq e^{n(\log R - \varepsilon)} = R^n e^{-n\varepsilon}$, where R is defined in (4.8). Fix some $c > 0$ and set $n_t := [ct]$. Note that $n_t \geq N_\varepsilon$ for any $t \geq t_\varepsilon := (N_\varepsilon + 1)/c$. For any $t \geq t_\varepsilon$,

$$a^{n_t} \theta^{n_t/2} U_{n_t}(t) = a^{n_t} \theta^{n_t/2} U_{n_t}(1) t^{\frac{4-\alpha}{2}n_t} \geq (a\sqrt{\theta}R)^{n_t} \frac{1}{(n_t!)^{\frac{4-\alpha}{2}}} t^{\frac{4-\alpha}{2}n_t} e^{-n_t\varepsilon}. \quad (6.20) \quad \boxed{\text{E:Asy1}}$$

Since $\lim_{t \rightarrow \infty} t^{-1} n_t = c$, we deduce that

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log (a^{n_t} \theta^{n_t/2} U_{n_t}(t)) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log (a^{n_t} \theta^{n_t/2} U_{n_t}(t)) =: I(n_t). \quad (6.21) \text{ ?E_-:Asy2?}$$

Now by (6.20), we see that

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left((a\sqrt{\theta}R)^{n_t} \frac{1}{(n_t!)^{\frac{4-\alpha}{2}}} t^{\frac{4-\alpha}{2} n_t} \right) - c\varepsilon \\ &= c \log(a\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(\left(\frac{t}{n_t} \right)^{\frac{4-\alpha}{2} n_t} \frac{n_t^{\frac{4-\alpha}{2} n_t}}{(n_t!)^{\frac{4-\alpha}{2}}} \right) - c\varepsilon \\ &= c \log(a\sqrt{\theta}R) - c \frac{4-\alpha}{2} \log c + c \frac{4-\alpha}{2} \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \frac{n_t^{n_t}}{n_t!} - c\varepsilon \\ &= c \log(a\sqrt{\theta}R) - c \frac{4-\alpha}{2} \log c + c \frac{4-\alpha}{2} - c\varepsilon, \end{aligned}$$

where the last equality is due to Stirling's formula. Letting $\varepsilon \rightarrow 0$ followed by optimizing c (the optimal c is $(a\sqrt{\theta}R)^{2/(4-\alpha)}$) proves (6.19). $\square$

We are now ready to give the proof of the desired lower bound:

Proof of Theorem 6.1. We return to Proposition 6.5. In relation (6.8), we take the supremum over all $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = a > 0$. Using the fact that $W_n(t, \phi) = a^n W_n(t, \phi/a)$ for any $\phi \in L^2(\mu)$ and the nonnegativity in (6.10), for any $c > 0$, we have that

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp \left\{ -\frac{1}{2}(q-1)t_p^\beta a^2 \right\} \sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=a} \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &= \exp \left\{ -\frac{1}{2}(q-1)t_p^\beta a^2 \right\} \sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=1} \sum_{n \geq 0} a^n \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &\geq \exp \left\{ -\frac{1}{2}(q-1)t_p^\beta a^2 \right\} \sup_{f \in \mathcal{H}_+; \|f\|_{\mathcal{H}}=1} a^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f), \end{aligned}$$

where $n_t = [c t_p^\beta]$. By choosing c to be the constant in Lemma 6.9, we see that

$$\liminf_{t_p \rightarrow \infty} \frac{1}{t_p^\beta} \log \|u(t, 0)\|_p \geq -\frac{q-1}{2} a^2 + (a\sqrt{\theta})^{\frac{2}{4-\alpha}} 2^{-\frac{\alpha}{2(4-\alpha)}} \mathcal{M}^{1/2} =: h(a).$$

We now maximize over $a > 0$. More precisely, let $b_1 = \frac{q-1}{2}$ and $b_2 = \theta^{\frac{1}{4-\alpha}} 2^{-\frac{\alpha}{2(4-\alpha)}} \mathcal{M}^{1/2}$. The maximum of the function $h(a) = -b_1 a^2 + b_2 a^{\frac{2}{4-\alpha}}$, $a > 0$ is attained at the point $a^* = \left(\frac{b_2}{b_1(4-\alpha)} \right)^{\frac{4-\alpha}{2(3-\alpha)}}$ and the maximum value of h is:

$$h(a^*) = (3-\alpha)(4-\alpha)^{-\frac{4-\alpha}{3-\alpha}} b_1^{-\frac{1}{3-\alpha}} b_2^{\frac{4-\alpha}{3-\alpha}}.$$

Replacing b_1 and b_2 by their corresponding values and noticing that $q-1 = 1/(p-1)$, we see that $h(a^*)$ coincides with the left-hand-side of (6.1). $\square$

7 Extension to other equations

?{S:Ext}? Our methods for studying the stochastic wave equation are quite general. With only some slight efforts, one can extend our results to the stochastic wave equation (1.1) to the following more general fractional stochastic partial differential equations (SPDE):

$$\begin{cases} \left(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2} \right) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right], & t > 0, x \in \mathbb{R}^d, \\ u(0, \cdot) = 1 & \text{if } b \in (0, 1], \\ u(0, \cdot) = 1, \quad \frac{\partial}{\partial t} u(0, \cdot) = 0 & \text{if } b \in (1, 2), \end{cases} \quad (7.1) \quad \boxed{\text{E2:SDPE}}$$

with $a \in (0, 2]$, $b \in (0, 2)$, $\nu > 0$, $r > 0$, and $\theta > 0$. In this equation, $(-\Delta)^{a/2}$ refers to the fractional Laplacian, the noise $\dot{W}$ is the same as that in (1.1), ∂_t^b denotes the *Caputo fractional differential operator*:

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^{(m)}(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

and I_t^r is the *Riemann-Liouville fractional integral* of order $r > 0$:

$$I_t^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) ds, \quad \text{for } t > 0,$$

with the convention $I_t^0 = \text{Id}$ (the identity operator). We refer to [8, 17, 18] for more details of these fractional differential operators. The SPDE (7.1) when $\dot{W}(x)$ is replaced by the space-time white noise $\dot{W}(t, x)$ is studied in Chen et al [5] for a general global Lipschitz $\rho(u)$ (instead of $\sqrt{\theta}u$) and for general initial conditions.

Following the notation in Chen et al [5], the fundamental solutions of (7.1) consist of a triplet:

$$\{Z_{a,b,\nu,d}(t, x), Z_{a,b,\nu,d}^*(t, x), Y_{a,b,r,\nu,d}(t, x) : t > 0, x \in \mathbb{R}^d\}$$

Since we are interested in the constant one initial position (and zero initial velocity when $b > 1$), by Theorem 4.1, *ibid.*, the corresponding solution to the homogeneous equation (i.e., the solution when there is no driven force) is equal to the constant one². Hence, by setting $G(t, x) = Y_{a,b,r,\nu,d}(t, x)$, (7.1) can be written in the following integral form:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t, x) u(s, y) W(\delta y) \right) ds. \quad (7.2) \quad \boxed{\text{E2:SPDEInt}}$$

In this setting, we need the notion $\mathcal{F}_a$ for $a \in (0, 2]$ as in Bass et al [3], namely,

$$\mathcal{F}_a := \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\},$$

²Indeed, from eq. (1.5) of [5], one only needs to check $\mathcal{F}Z(t, \cdot)(0) = 1$ when $b \in (0, 1]$ and $\mathcal{F}Z^*(t, \cdot)(0) = 1$ when $b \in (1, 2]$, which are clear from eq.'s (4.7) and (4.9), *ibid.*, respectively.

where $H^a(\mathbb{R}^d)$ is the Sobolev space of the (fractional) order a with $H^2(\mathbb{R}^d) = W^{1,2}(\mathbb{R}^d)$, and

$$\mathcal{E}_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi.$$

For any nonnegative definite measure f and $\theta > 0$, define

$$\mathcal{M}_a(f, \theta) = \sup_{g \in \mathcal{F}_d} \left\{ \langle g^2 * f, g^2 \rangle^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\}, \quad (7.3) \text{ ?E2:cM?}$$

which reduces to $\mathcal{M}(f, \theta)$ as in (1.8) when $a = 2$. By the same arguments as the proof of Lemma 2.3, one can show the following scaling property:

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1), \quad \text{for all } \Theta > 0 \text{ and } \theta > 0,$$

according to which, relation (2.8) becomes

$$\rho_a = 2^{-\alpha/a} \mathcal{M}_a^{(2a-\alpha)/a}.$$

The following assumption plays the role of the condition $d \leq 3$ in Theorem 1.6, since it guarantees the fundamental solution $G(t, x)$ is nonnegative; see Lemma 7.4 below.

Assumption C. Assume the parameters (a, b, r) and the dimension d for (7.1) satisfy one of the following four conditions (as illustrated in Figure 1):

- (1) $d \geq 1$, $b \in (0, 1)$, $a \in (0, 2]$, $r \geq 0$;
- (2) $d \geq 1$, $b = 1$, $a \in (0, 2]$, $r = 0$ or $r > 1$;
- (3) $1 \leq d \leq 3$, $1 < b < a \leq 2$, $r > 0$;
- (4) $1 \leq d \leq 3$, $1 < b = a < 2$, $r > \frac{d+3}{2} - b$.

Now we are ready to state our extension theorem as follows:

Theorem 7.1. Under Assumptions A and C, the SDPE (7.1) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$, $t > 0$ and $x \in \mathbb{R}^d$ if

$$\alpha < \min \left(\frac{a}{b} (2[b+r] - 1), 2a, d \right). \quad (7.4) \text{ E2:iffcond}$$

Moreover, under condition (7.4), by setting

$$t_p := (p-1)^{\frac{a}{2[b+r]a-b\alpha}} t \quad \text{and} \quad \beta := \frac{2a[b+r] - b\alpha}{a(2[b+r] - 1) - b\alpha}, \quad (7.5) \text{ ?E2:tp-beta?}$$

it holds that

$$\begin{aligned} \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &= \theta^{\frac{a}{a(2[b+r]-1)-b\alpha}} \nu^{-\frac{\alpha}{a(2[b+r]-1)-b\alpha}} \\ &\times \left(2[b+r] - \frac{b}{a}\alpha - 1 \right) \left(\frac{2\mathcal{M}_a^{1/a}}{2[b+r] - \frac{b}{a}\alpha} \right)^{\frac{2a[b+r]-b\alpha}{a(2[b+r]-1)-b\alpha}}. \end{aligned} \quad (7.6) \text{ E2:Main}$$

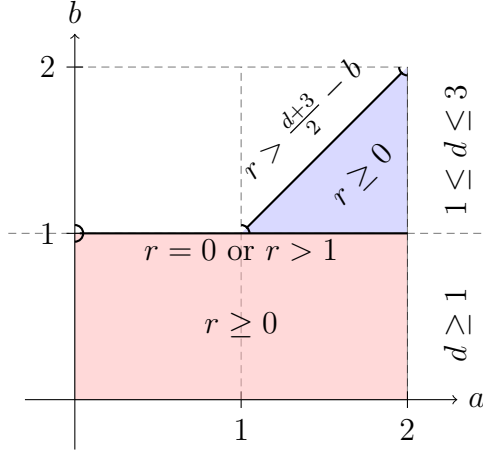


Figure 1: Assumption **C** — some sufficient conditions for $G(t,)$ to be nonnegative.

F:Nonnegative

In particular, by freezing an arbitrary $p \geq 2$ in (7.6), we have that

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{-\beta} \log \|u(t, x)\|_p &= p([p-1]\theta)^{\frac{a}{a(2[b+r]-1)-b\alpha}} \nu^{-\frac{\alpha}{a(2[b+r]-1)-b\alpha}} \\ &\times \left(2[b+r] - \frac{b}{a}\alpha - 1\right) \left(\frac{2\mathcal{M}_a^{1/a}}{2[b+r] - \frac{b}{a}\alpha}\right)^{\frac{2a[b+r]-b\alpha}{a(2[b+r]-1)-b\alpha}} \end{aligned} \quad (7.7) \text{ ?E2:tMain?}$$

and by freezing an arbitrary $t > 0$ in (7.6), we have that

$$\begin{aligned} \lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) &= t^{\frac{2a[b+r]-b\alpha}{a(2[b+r]-1)-b\alpha}} \theta^{\frac{a}{a(2[b+r]-1)-b\alpha}} \nu^{-\frac{\alpha}{a(2[b+r]-1)-b\alpha}} \\ &\times \left(2[b+r] - \frac{b}{a}\alpha - 1\right) \left(\frac{2\mathcal{M}_a^{1/a}}{2[b+r] - \frac{b}{a}\alpha}\right)^{\frac{2a[b+r]-b\alpha}{a(2[b+r]-1)-b\alpha}}. \end{aligned} \quad (7.8) \text{ ?E2:pMain?}$$

SpeicalCases? **Remark 7.2.** Condition (7.4) reduces to

$$\begin{cases} \alpha < 3 \wedge d, & \text{for SWE, i.e., } a = b = 2 \text{ and } r = 0; \text{ and} \\ \alpha < 2 \wedge d, & \text{for SHE, i.e., } a = 2, b = 1 \text{ and } r = 0. \end{cases}$$

Remark 7.3. Formally, the white noise case corresponds the case when $\alpha = d$. The solvability condition (7.4) suggests that this is possible only when

$$d < \min\left(2a, \frac{a}{b}(2[b+r]-1)\right) = 2a + \frac{a}{b} \min(2r-1, 0),$$

which coincides the condition obtained in Chen et al [5]; see (1.9), ibid. Since $a \in (0, 2]$, the highest dimension for which one can make sense of solution using white noise is $d = 3$.

By checking studying the wave equation case, we see that for a general stochastic integral equation (7.2), if only if one has the three properties list in the Lemma 7.4 below, then one can apply the same arugments to derive the solvability condition and the exact moment asymptotics.

$\langle \text{L2:G} \rangle$ **Lemma 7.4.** For any $a \in (0, 2]$, $b \in (0, 2)$, $\nu > 0$, $r > 0$, and $d \geq 1$, the fundamental solution $G(t, x)$, i.e., $Y_{a,b,r,\nu,d}(t, x)$ in [5], satisfies the following properties:

(i) The scaling property in the following equivalent forms: for any $c > 0$,

$$\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi) \quad \text{or} \quad \mathcal{F}G(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}G(t, \cdot)\left(c^{\frac{b}{a}}\xi\right).$$

(ii) The Fourier-Laplace transform of G is equal to

$$\mathcal{L}\mathcal{F}G(s, \xi) = \frac{s^{-r}}{s^b + 2^{-1}\nu|\xi|^a}, \quad \text{for any } \xi \in \mathbb{C} \text{ and } \Re(s) > -2^{-1/b}\nu^{1/b}|\xi|^{a/b}.$$

(iii) Under Assumption **C**, $G(t, x)$ is a nonnegative and smooth function for all $t > 0$ and $x \neq 0$.

Proof. Property (i) is due to (4.8) of Chen et al [5]. Property (ii) is proved in the proof of Theorem 4.1, *ibid*. The nonnegativity in part (iii) is proved in Theorem 4.6, *ibid*. The claim of G being smooth for $t \neq 0$ and $x \neq 0$ is a consequence of the explicit formula of G in (4.5), *ibid*. $\square$

Sketch of proof of Theorem 7.1. The proof consists of the following four steps:

Step 1 (Solvability). With the scaling property in part (i) of Lemma 7.4, (3.16) becomes

$$\left\| \tilde{f}(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{(2[b+r]-\frac{b}{a}\alpha)n} \left\| \tilde{f}(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (7.9) \text{ ?E2:scaleTildeFn}$$

Because

$$\alpha < \frac{a}{b}(2[b+r]-1) \iff 2[b+r] - \frac{b}{a}\alpha > 1, \quad (7.10) \text{ E2:Cond}$$

condition (7.4) is sufficient to guarantee that for all $n \geq 1$,

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma\left(\left(2[b+r] - \frac{b}{a}\alpha\right)n + 1\right) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (7.11) \text{ ?E2:GammaLapInt?}$$

By checking the *Case 1* part of the proof of Lemma 3.6, one can apply part (ii) of Lemma 7.4 with $s = 1$ to see that (3.8) becomes

$$\begin{aligned} & \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt \\ & \leq 2^{(2[b+r]-\frac{b}{a}\alpha)n} \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + 2^{-1}\nu|\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n), \end{aligned}$$

where we have used the fact that G is nonnegative under our current assumption. Then by the scaling property of μ , we see that

$$\begin{aligned} \int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt &\leq \left(\frac{\nu}{2}\right)^{-(n\alpha)/a} 2^{(2[b+r]-\frac{b}{a}\alpha)n} \frac{1}{(n!)^2} \\ &\quad \times \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\xi_1) \dots \mu(d\xi_n). \end{aligned} \quad (7.12) \quad \text{?E2:UB-Laplace-n}$$

Hence, by the same arguments as those in Lemme 3.6,

$$\int_0^\infty e^{-t} \|\tilde{f}_n(\cdot, 0; t)\|_{\mathcal{H}^{\otimes n}}^2 dt \leq \left(\frac{\nu}{2}\right)^{-\frac{n\alpha}{a}} 2^{(2[b+r]-\frac{b}{a}\alpha)n} C_{\mu,a}^n =: \Theta^n,$$

with

$$C_{\mu,a} := \int_{\mathbb{R}^d} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) < \infty \iff \alpha < 2a. \quad (7.13) \quad \boxed{\text{E2:alpha<2a}}$$

Therefore,

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} \theta^n n! t^{(2[b+r]-\frac{b}{a}\alpha)n} \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} \frac{\theta^n n! t^{(2[b+r]-\frac{b}{a}\alpha)n}}{\Gamma((2[b+r]-\frac{b}{a}\alpha)n + 1)} \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt \quad (7.14) \quad \text{?E2:Eusq?} \\ &\leq \sum_{n \geq 0} \frac{n!}{\Gamma((2[b+r]-\frac{b}{a}\alpha)n + 1)} \theta^n \Theta^n t^{(2[b+r]-\frac{b}{a}\alpha)n}, \end{aligned}$$

which is finite for all $t > 0$ thanks to condition in (7.10). Notice that Assumption A requires $\alpha < d$. Finally, an application of Theorem 2.2 proves the existence and uniqueness of the solution.

Step 2 (Case $p = 2$). The aim of this step is to show the corresponding Theorem 4.1. By the same arguments as those in the proof of Lemma 4.2, one can show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma \left(\left(2[b+r] - \frac{b}{a}\alpha \right) n + 1 \right) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left(\nu^{-\alpha/a} 2^{2[b+r]-\frac{b}{a}\alpha} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right),$$

or equivalently,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2[b+r]-\frac{b}{a}\alpha} \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left(\nu^{-\alpha/a} \left(\frac{2}{2[b+r]-\frac{b}{a}\alpha} \right)^{2[b+r]-\frac{b}{a}\alpha} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right).$$

With this limit, one can carry out the same arguments as those in the proof of Theorem 4.1 to see that (4.10) is still true with β , $3 - \alpha$, and R replaced by

$$\frac{2[b+r] - \frac{b}{a}\alpha}{2[b+r] - \frac{b}{a}\alpha - 1}, \quad 2[b+r] - \frac{b}{a}\alpha - 1, \quad \text{and} \quad \nu^{-\alpha/a} \left(\frac{2}{2[b+r] - \frac{b}{a}\alpha} \right)^{2[b+r] - \frac{b}{a}\alpha} \mathcal{M}_a^{\frac{2a-\alpha}{a}}.$$

Therefore,

$$\begin{aligned} \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E} (|u(t, x)|^2) &= \theta^{\frac{a}{a(2[b+r]-1)-b\alpha}} \nu^{-\frac{\alpha}{a(2[b+r]-1)-b\alpha}} \\ &\times \left(2[b+r] - \frac{b}{a}\alpha - 1 \right) \left(\frac{2 \mathcal{M}_a^{1/a}}{2[b+r] - \frac{b}{a}\alpha} \right)^\beta. \end{aligned} \quad (7.15) \quad \boxed{\text{E2:p2}}$$

Step 3 (Upper bound in (7.6)). Thanks to (7.15), one can apply for the hypercontractivity property as the proof of Theorem 5.1 to see that $\limsup_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p$ is bounded from above by the right-hand side of (7.6).

Step 4 (lower bound in (7.6)). The proof of the lower bound is similar to that of Theorem 6.1 in Section 6. We need to make the following changes:

- (i) We need to replace the condition $0 < \alpha < d \leq 3$ by Assumption C. Indeed, the condition $0 < \alpha < d$ has already been imposed in Assumption A and the condition $d \leq 3$ is to make sure the fundamental solution G is nonnegative, which is guaranteed by Assumption C (see Lemma 7.4).
- (ii) Similarly, we need to replace the condition $d < 3$ by Assumption C. Recall that by Lemma 7.4, Assumption C ensures that G is not only a nonnegative function, but also a smooth function for $t > 0$ and $x \neq 0$. We won't see measures as the fundamental solution for the wave equation in dimension three.
- (iii) As proved in the Step 1, the counter part of the condition $\alpha < 3$ in Theorem 1.6 is (7.4). Hence, we need to replace all conditions $\alpha < 3$ by (7.4).
- (iv) We need to replace the condition $\alpha < 4$ by $\alpha < 2a$; see (7.13).
- (v) In the proof of Theorem 6.6, there is no need to introduce the space $\mathcal{H}_+$ since the fundamental solutions are... I still need to elaborate here with some ...

□

A Exponential behaviour of power series

In this part we examine the asymptotic behaviour of some power series, measured on the exponential scale. The first result is a useful tool for comparing two power series, the proof of which follows the same arguments as those in the proof of Theorem 1.3 of Balan and Song [1].

$\langle \text{a-b-lemma} \rangle$ **Lemma A.1.** *If $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$ are two sequences of positive real numbers such that*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \frac{b_n}{a_n} \geq \log \rho \quad \text{and} \quad \liminf_{t \rightarrow \infty} \frac{1}{t^p} \log \sum_{n \geq 0} a_n t^n \geq A$$

for some $\rho > 0, p > 0$ and $A \in \mathbb{R}$, then $\liminf_{t \rightarrow \infty} \frac{1}{t^p} \log \sum_{n \geq 0} b_n t^n \geq \rho^p A$. The same statement remains valid if we replace $(\liminf, \geq)$ by $(\limsup, \leq)$ or $(\lim, =)$.

The next lemma is an application of the previous result. It says that, if $R_n^{1/n} \approx R$ as $n \rightarrow \infty$, then we can replace R_n by R^n when deriving the exponential asymptotic behaviour of the power series $\sum_{n \geq 0} x_n R_n t^n$.

$\langle \text{Rn-compare} \rangle$ **Lemma A.2.** *If $(R_n)_{n \geq 0}$ and $(x_n)_{n \geq 0}$ are two sequences of positive real numbers such that*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log R_n \geq \log R \quad \text{and} \quad \liminf_{t \rightarrow \infty} \frac{1}{t^p} \log \sum_{n \geq 0} x_n R^n t^n \geq A$$

for some $R > 0, p > 0$ and $A \in \mathbb{R}$, then $\liminf_{t \rightarrow \infty} t^{-p} \log \sum_{n \geq 0} x_n R_n t^n \geq A$. The same statement remains valid if we replace $(\liminf, \geq)$ by $(\limsup, \leq)$ or $(\lim, =)$.

Proof. The lemma is proved by an application of Lemma A.1 with $a_n = x_n R^n$ and $b_n = x_n R_n$, and the fact that $\liminf_{n \rightarrow \infty} \frac{1}{n} \log \frac{R_n}{R^n} = \liminf_{n \rightarrow \infty} \frac{1}{n} \log R_n - \log R \geq 0$. $\square$

The next lemma is about the asymptotic property of the *Mittag-Leffler* function [17].

$\langle \text{L:Gamma} \rangle$ **Lemma A.3.** *For any $\gamma > 0$, $\lim_{t \rightarrow \infty} t^{-1/\gamma} \log \sum_{n \geq 0} (n!)^{-\gamma} t^n = \gamma$.*

Proof. We start by looking at the similar result with $(n!)^\gamma$ replaced by $\Gamma(\gamma n + 1)$:

$$\lim_{t \rightarrow \infty} \frac{1}{t^{1/\gamma}} \sum_{n \geq 0} \frac{t^n}{\Gamma(\gamma n + 1)} = \lim_{t \rightarrow \infty} \frac{1}{t} \sum_{n \geq 0} \frac{t^{\gamma n}}{\Gamma(\gamma n + 1)} = 1,$$

where the last equality is due to the asymptotic property of the Mittag-Leffler function; see Theorem 1.3 on p. 32 and Theorem 1.7 on p. 35 of [17]. This lemma is prove by an application of (4.3) and Lemma A.1 with $a_n = \frac{1}{\Gamma(\gamma n + 1)}$, $b_n = \frac{1}{(n!)^\gamma}$ and $p = 1/\gamma$. $\square$

B Moment comparison using hypercontractivity

This section gives the moment comparison result of Khoa Lê [14]. This result was stated in [14] for the Parabolic Anderson Model with a special Gaussian noise, but Lê's proof is in fact valid in a much more general case. We present this proof here, including some of the details which are missing from [14].

Let $W = \{W(\varphi); \varphi \in \mathcal{H}\}$ be an isonormal Gaussian process, associated to a Hilbert space $\mathcal{H}$. Assume that either one of the following conditions hold:

- (i) $\mathcal{H}$ consists of functions (or distributions) on $\mathbb{R}_+ \times \mathbb{R}^d$, i.e. W is *time-dependent*; and
- (ii) $\mathcal{H}$ consists of functions (or distributions) on $\mathbb{R}^d$, i.e. W is *time-independent*.

Let $\mathcal{L}$ be a second-order pseudo-differential operator of $\mathbb{R}_+ \times \mathbb{R}^d$ and u_θ be the solution of the SPDE:

$$\mathcal{L}u(t, x) = \sqrt{\theta}u(t, x)\dot{W}, \quad t > 0, x \in \mathbb{R}^d \quad (\text{B.1}) \quad \boxed{\text{E:Leq}}$$

with (deterministic) initial condition. By definition, the (*mild Skorohod*) *solution* to (B.1) is an adapted square-integrable process $u_\theta = \{u_\theta(t, x); t > 0, x \in \mathbb{R}^d\}$ which satisfies

$$u_\theta(t, x) = J_0(t, x) + \sqrt{\theta} \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) u_\theta(s, y) W(\delta s, \delta y),$$

if the noise W is time-dependent, respectively

$$u_\theta(t, x) = J_0(t, x) + \sqrt{\theta} \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) u_\theta(s, y) W(\delta y) ds,$$

if the noise W is time-independent, provided that these integrals are well-defined. Here $W(\delta s, \delta y)$ (respectively $W(\delta y)$) denotes the Skorohod integral with respect to W , G is the fundamental solution of $\mathcal{L}$ on $\mathbb{R}_+ \times \mathbb{R}^d$, and J_0 is the solution of the deterministic equation $\mathcal{L}u = 0$ on $\mathbb{R}_+ \times \mathbb{R}^d$, with the same initial condition as (B.1).

$\langle \text{T:Khoa} \rangle$ **Theorem B.1.** *If for any $\theta > 0$, equation (B.1) has a unique solution $u_\theta = \{u_\theta(t, x); t > 0, x \in \mathbb{R}^d\}$ and $\mathbb{E}(|u_\theta(t, x)|^p) < \infty$ for any $t > 0, x \in \mathbb{R}^d$ and $p > 1$, then*

$$\left\| u_{\frac{p-1}{q-1}\theta}(t, x) \right\|_q \leq \|u_\theta(t, x)\|_p \quad \text{for any } 1 < p \leq q. \quad (\text{B.2}) \quad \boxed{\text{mom-comp}}$$

In particular,

$$\|u_\theta(t, x)\|_p \leq \|u_{(p-1)\theta}(t, x)\|_2 \quad \text{for any } p \geq 2.$$

Proof. The proof is based on Mehler's formula. First, we need to introduce the framework for this result, as presented in Section 1.4 of Nualart [16]. Suppose that W is defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where $\mathcal{F}$ is generated by W . Let $W' = \{W'(\varphi); \varphi \in \mathcal{H}\}$ be a copy of W , defined on $(\Omega', \mathcal{F}', \mathbb{P}')$, where $\mathcal{F}'$ is generated by W' . Using the projection maps, we redefine W and W' on $(\Omega \times \Omega', \mathcal{F} \times \mathcal{F}', \mathbb{P} \times \mathbb{P}')$, so that W and W' are independent.

For any $\tau > 0$ and $\varphi \in \mathcal{H}$, let

$$Z_\tau(\varphi) = e^{-\tau} W(\varphi) + \sqrt{1 - e^{-2\tau}} W'(\varphi).$$

Then $Z_\tau = \{Z_\tau(\varphi); \varphi \in \mathcal{H}\}$ is also an isonormal Gaussian process, defined on $\Omega \times \Omega'$.

For any $\mathcal{F}$ -measurable function $F : \Omega \rightarrow \mathbb{R}$, there exists a measurable map $\psi_F : \mathbb{R}^{\mathcal{H}} \rightarrow \mathbb{R}$ such that $F = \psi_F(W)$. Let $(T_\tau)_{\tau \geq 0}$ be the Ornstein-Uhlenbeck semigroup on $L^2(\Omega)$. By Mehler's formula, for any $\tau > 0$ and $F \in L^2(\Omega)$

$$T_\tau(F) = \mathbb{E}'[\psi_F(Z_\tau)] \quad \mathbb{P}\text{-a.s.},$$

where $\mathbb{E}'$ denotes the expectation with respect to $\mathbb{P}'$. We apply this formula to $F = u_\theta(t, x)$. We denote $\psi_{u_\theta(t, x)} = \psi_{\theta, t, x}$. Then $u_\theta(t, x) = \psi_{\theta, t, x}(W)$ and

$$T_\tau(u_\theta(t, x)) = \mathbb{E}'[\psi_{\theta, t, x}(Z_\tau)] \quad \mathbb{P}\text{-a.s.} \quad (\text{B.3}) \quad \boxed{\text{Mehler}}$$

Since $\{\psi_{\theta,t,x}(W); t \geq 0, x \in \mathbb{R}^d\}$ is a solution to (B.1), $\{\psi_{\theta,t,x}(Z_\tau); t \geq 0, x \in \mathbb{R}^d\}$ is a solution of

$$\mathcal{L}u(t, x) = \sqrt{\theta}u(t, x)\dot{Z}_\tau, \quad t > 0, x \in \mathbb{R}^d, \quad (\text{B.4}) \quad \boxed{\text{E:Leq2}}$$

with the same initial condition as (B.1). Denote $u_{\tau,\theta}(t, x) := \psi_{\theta,t,x}(Z_\tau)$. Then (B.3) becomes:

$$T_\tau(u_\theta(t, x)) = \mathbb{E}'[u_{\tau,\theta}(t, x)] \quad \mathbb{P}\text{-a.s.} \quad (\text{B.5}) \quad \boxed{\text{Mehler2}}$$

Assume that the noise W is time-dependent. (The case of the time-independent noise is similar.) Since $\{u_{\tau,\theta}(t, x); t \geq 0, x \in \mathbb{R}^d\}$ is a solution of (B.4) with the same initial condition as (B.1), we see that

$$\begin{aligned} u_{\tau,\theta}(t, x) &= J_0(t, x) + \sqrt{\theta} \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) u_{\tau,\theta}(s, y) Z_\tau(\delta s, \delta y) \\ &= J_0(t, x) + \sqrt{e^{-2\tau}\theta} \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) u_{\tau,\theta}(s, y) W(\delta s, \delta y) \\ &\quad + \sqrt{\theta(1-e^{-2\tau})} \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) u_{\tau,\theta}(s, y) W'(\delta s, \delta y). \end{aligned}$$

We take expectation with respect to $\mathbb{P}'$. The third term on the right-hand side above disappears since the Skorohod integral has zero mean. Since W and W' are independent, the expectation with respect to $\mathbb{P}'$ commutes with the Skorohod integral with respect to W and hence

$$\mathbb{E}'[u_{\tau,\theta}(t, x)] = J_0(t, x) + \sqrt{e^{-2\tau}\theta} \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) \mathbb{E}'[u_{\tau,\theta}(s, y)] W(\delta s, \delta y).$$

This proves that the process $\{u_{\tau,\theta}(t, x); t \geq 0, x \in \mathbb{R}^d\}$ is a solution of

$$\mathcal{L}u = \sqrt{e^{-2\tau}\theta}u\dot{W}, \quad t > 0, x \in \mathbb{R}^d$$

with the same initial condition as (B.1). Since this equation has the *unique* solution $u_{e^{-2\tau}\theta}$, we conclude that for any $t > 0$ and $x \in \mathbb{R}^d$,

$$\mathbb{E}'[u_{\tau,\theta}(t, x)] = u_{e^{-2\tau}\theta}(t, x) \quad \mathbb{P}\text{-a.s.}$$

Combining this with (B.5), we obtain that for any $\tau > 0, t > 0$ and $x \in \mathbb{R}^d$,

$$T_\tau(u_\theta(t, x)) = u_{e^{-2\tau}\theta}(t, x) \quad \mathbb{P}\text{-a.s.} \quad (\text{B.6}) \quad \boxed{\text{Mehler3}}$$

By the hypercontractivity of the Ornstein-Uhlenbeck semigroup (see, e.g., Theorem 5.1 of Nualart [15] or Theorem 1.4.1 of Janson [16]), for any $\tau > 0, p > 1$ and $F \in L^p(\Omega)$,

$$\|T_\tau F\|_{q(\tau)} \leq \|F\|_p$$

where $q(\tau) = e^{2\tau}(p-1) + 1$. We apply this to $F = u_\theta(t, x)$. Using (B.6), we obtain:

$$\|u_{e^{-2\tau}\theta}(t, x)\|_{q(\tau)} \leq \|u_\theta(t, x)\|_p.$$

For fixed $1 < p \leq q$, choose $\tau > 0$ such that $q(\tau) = q$. Then $e^{-2\tau} = \frac{p-1}{q-1}$ and (B.2) follows. $\square$

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